

DESIGN, SIMULATION, AND DEMONSTRATION OF GLOBAL NAVIGATION SATELLITE SYSTEM (GNSS)-BASED RADAR ALTIMETER



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August 2021

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A thesis submitted to the
Institute of Space Technology
In the partial fulfillment of the requirements
For the degree of Master of Science in
(Global Navigation Satellite System)

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System (GNSS) based Radar Altimeter

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ACKNOWLEDGEMENT

I would like to thank Almighty ALLAH for His countless blessings for giving me knowledge and direction in completing my research. I admit all the support and guidance that I have received during my research work and express my most sincere appreciation to my thesis supervisor: Dr. Najam Abbas Naqvi, who was always forthcoming in making my thesis a success. I also acknowledge the keen interest of Dr. Muhammad Usman in my experimental effort due to which my analysis of the thesis was successful.

A special thanks to Dr. Umer Iqbal Bhatti for giving his valuable inputs and suggestions. During my sequence of work without the guidance and inspiration of Dr. Zainab Farooq, I would not have achieved success in my thesis.

And finally, I would like to thank my parents, who always supported me during my studies and research work. In the end, my special gratitude towards my colleagues, for their helps to make this thesis a success.

CERTIFICATE

This is to certify that the research work described in this MS Thesis is the original work of the author and has been carried out under my direct supervision. I have personally gone through all the data/results/materials reported in the manuscript and certify their correctness/authenticity. I further certify that material included in this thesis is not plagiarized and has not been used in part or full in a manuscript already submitted or in the process of submission in partial/complete fulfillment of the award of any other degree from any institution. I also certify that the thesis has been prepared under my supervision according to the prescribed format and I endorse its evaluation for the award of Master of Science in Global Navigation Satellite System (GNSS) through the official procedures of the Institute.

Dr. Najam Abbas Naqvi

DEDICATION

I dedicate this research thesis to my beloved parents, my wife, and my sister because their prayers are always with me.

ABSTRACT

The Global Navigation Satellite System (GNSS) has many applications beyond the standard applications of positioning, timing, and navigation. The remote sensing application is based on the refractive and reflective properties of direct and reflected GNSS signals. The refractivity is used in GNSS Radio Occultation (RO) while reflectivity is used in GNSS reflectivity or GNSS R. GNSS-based radar altimetry is one of the secondary applications that use the code delay, carrier phase, and signal-to-noise ratio (SNR) for height measurement. In this research endeavor, the idea of GNSS R-based altimetry is demonstrated using an experimental setup.

GPS L1 signal was generated for a specific constellation using Aeroflex® Global Positioning System (GPS) simulator equipment. The experiment was performed in an enclosed environment and in front of a concrete wall. The signal was transmitted using a Right Hand Circularly Polarized (RHCP) helical antenna, as carried out in an actual GPS satellite. The signal was reflected from a target that was a cemented wall and then both the direct and the reflected signals were received with the help of self-made RHCP and LHCP antennas, respectively. The data was converted to Intermediate Frequency (IF) using a four-channel Radio Frequency (RF) front end and then both direct and reflected signals were sampled at 53MHz and then stored in a binary format for post-processing. The acquisition was performed for both signals using customized software developed in Matlab® and then the correlation peaks were compared in terms of chip delay and then the distance was estimated using the chip difference. Through the various experiments, it was demonstrated that the actual distance between the RHCP transmits and LHCP receive antenna roughly corresponded to the chip delay and thus validated the idea that the reflected GNSS signals could be used for passive radar altimetry.

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ABBREVIATIONS AND ACRONYMS

GNSS	Global Navigation satellite systems
RO	Radio Occultation
SNR	Signal-to-noise Ratio
GPS	Global positioning system
RHCP	Right handed circular polarized
LHCP	Left handed circular polarized
IF	Intermediate Frequency
QZSS	Quazi Zenith Satellite System
IRNSS	Indian Regional Navigation satellite system
GLONASS	Globalnaya Navigazionnaya Sputnikovaya Sistema
IOV	In orbit validation
FOC	Full operational capability
IGSO	Inclined geosynchronous orbit
MEO	Medium Earth orbit
NavIC	Navigation with Indian constellation
LEO	Low Earth Orbit
ADC	Analogue to Digital Converter
C/A	Coarse Acquisition
C/No	Carrier to Noise ration
FM-CW	Frequency modulated continuous wave
WAAS	Wide Area Augmentation System
CORS	Continuously Operating Reference Stations
NOAA	Global Differential GPS
GDGPS	Global Differential GPS
NASA	National Aeronautics and space Administration
IGS	International GNSS Service
EGNOS	European Geostationary Navigation Overlay Service
MSAS	Multifunctional satellite Augmentation system
GAGAN	GPS Aided GEO augmentation navigation
SDCM	System for Differential Correction & Monitoring

1 INTRODUCTION

1.1 Background

The GNSS now has the full operational capability and has many applications in the form of position, Timing, and Navigation. GNSS multipath signals are primary error source in all these applications but on the other hand, these reflected signals provide additional opportunity for remote sensing applications over last 3 decades. Remote sensing is a technique to extract information about land features using reflected electromagnetic waves. The reflected wave carries a unique signature for a certain land cover. Satellite remote sensing and radar altimetry are also types of remote sensing which use microwaves to measure the height from sea level.

Flying machines use different methods for altimetry. The basic principle is that the speed of electromagnetic waves is equals to the speed of light and by calculating travel time difference between signal transmission and receiving we can calculate the distance in terms of height. The GNSS is a signal of opportunity for remote sensing and altimetry as more than 100 transmitters are available globally, for any time and any weather with known altitude. The most important difference between radar altimeter and GNSS altimeter is of the non-cooperative biostatic radar of opportunity. Multiple satellites are always available globally and provide signals for free. As GNSS signals have unique signatures which help to acquire these weak signals. The reflected signals from land cover and any surface have to cover the extra distance from the direct signal from a satellite which introduces a delay between these two. The calculation of code delay in terms of distance turns into a Bistatic radar altimeter. Radar altimeter measures the vertical distance along a single track while GNSS altimeter can perform multiple experiments for altimetry using different ground tracks. The correlation functions for both signals are calculated which results in the delay between both in terms of chips. The resolution of the altimeter depends upon the sampling frequency of front. Better the resolution can be achieved using higher frequency.

In my project we provide an idea for design and simulate the GNSS based altimeter using GPS L1 signals. Both direct and reflected signals are compared to retrieve the chip difference in terms of delay. This type of sensor will provide altimetry services permanent in all weather conditions; it will be economical and ergonomic as compared to conventional altitude meters available in the market. The GNSS has many secondary applications beyond the primary application of timing, positioning, and Navigation.. However, developments have revealed that these reflected signals can be used for certain calculations regarding land features of different land covers .i.e. ocean surface parameters, wind velocity, and other land features When the GNSS signal reflect from any surface it scatters in all directions but the maximum power is distributed along the forward direction of the specular point. The signals when reflected have unique code

delay and Doppler shift. The code delay can be used to achieve various information for altimetry.

The Objective of the research work is to receive and analyze these weak reflected signals using an LHCP antenna to extract the required information. The research work will lead towards a cheap solution for GNSS-Based radar altimeter which can be used at any time or even in bad weather storms likewise barometric altimeters with lots of conditional limitations.

1.2 Problem statement

From last three decades, It has been are continuously attempted to find out economical methods for design of altimeter using the un-directional waves of GNSS. My interest to write this thesis is similar to find out best solution for this problem. This research work will provide algorithm for altimetry and may be used for other GNSS applications. The developed algorithm will perform a comparison of direct and reflected signals correlation plots.

1.3 Objective

The focus and goal of this study are to calculate the distance using a comparison of the correlation waveforms in terms of chip delay between reflected and direct signals. The code delay is extracted in terms of code chips for distance calculation. The purpose of this study can be further described as following

Design of cheap hardware for GNSS signal-based altimetry

1. Development of an algorithm for processing of reflected signals
2. Understanding scenarios for better reception of reflected signals

1.4 Research Limitations

To receive a weak GNSS reflected signal, we required an LHCP antenna with precise measurements, while the available antenna was custom-made. We should arrange a commercial antenna with a stable phase center so that more precise results could be obtained.

The sampling frequency of the available front end was 53 MHz which limited the bit resolution to 5.7m. This can be also be increased by using the front end with a higher sampling frequency.

The number of experiments could be performed in an open atmosphere with different delays. This would increase the number of samples leading towards a better conclusion. But unfortunately, the availability of the equipment was limited to the GNSS lab.

The Equipment was available for a very short period so we were unable to perform in multiple scenarios. The constellation and a number of available bands also restricted the research in terms of conclusion and analysis. In the future, if someone performs without these constraints, the results will be more fruitful and conclusive.

1.5 Thesis Outline

This research work has been described in six (06) chapters. A brief overview regarding content regarding next chapters is following

Chapter No 2 :(GNSS)

This chapter describes a brief description about 1 Global; regional navigation systems.it also contains information about different Augmentation systems

Chapter No 3 :(GNSS SIGNAL ACQUISITION AND ALTIMETRY)

For a better understanding of GNSS reflected signals, it is mandatory to be familiar with GNSS basic concepts and signal characteristics. This chapter furnished a description of GNSS reflected signals for remote sensing and altimetry applications. This chapter also contains a detailed description of GNSS signal acquisition methods and how to differentiate acquisition plots for the measurement of extra path delay in reflected signals. It also explains GNSS Based Radar altimetry

Chapter No 4 :(DATA ACQUISITION AND METHODOLOGY)

This Chapter provides an explanation of used hardware for GNSS data acquisition with the help of Custom made LHCP antenna and 4 channel Front ends. It also describes the algorithm for signal acquisition and comparison of both signals in terms of chip delay.

Chapter No 5 :(RESULTS AND ANALYSIS)

This chapter focuses on the results deduced from practical experiments and simulations. The results are supported by different Matlab® diagrams and then these are compared, discussed, and analyzed

Chapter No 6 :(CONCLUSION)

This final chapter provides the analysis and direction for future work recommendations.

2 GLOBAL NAVIGATION SATELLITE SYSTEM

Christopher was not the only person who proposed that the earth was round, but Eratosthenes around 240BE also calculated the earth's radius with surprising accuracy. Columbus navigated the new world of America using dead reckoning, a technique of current position calculations using the previous one. For example, if I start my Journey from a Known point to the west direction with the speed of 10 Km/h, after 2 hours, I will be 20 km westward from the reference. In 1714, the British people took the initiative for the development of many navigation techniques and equipment .i.e. shipborne chronometers. In the 1900s, these chronometers were regulated by radio signals. In the 1940s, ships started to triangulate their position using LORAN signals. In 1957, Sputnik satellite was launched with which many of the navigational problems started becoming historical footstones.[1]

GNSS is a spaced-based positioning, navigation and timing system that is used for prediction of coordinates in space, concerning velocity and time, to assist in navigation and positioning. The service is provided by various global constellations such as Americans GPS, Russian GLONASS, Europeans Union Galilei, Chinese Beidou, and regional constellations including compass ,Japanese QZSS), and IRNSS from Indians.[2]

The advent of the GNSS changed navigation techniques significantly. First, in 1970, the US department of defense launched a Global Positing System for military purposes but later on, with the improvement in accuracy and precision, it became publicly available for surveying, mapping, and even our daily routine works [1]. The Following GNSS systems are operational nowadays.

1. GPS
2. GLONASS
3. BeiDou
4. Galileo
5. IRNSS
6. QZSS

2.1 Global Positioning System

As of June 2021, this constellation is consisting of 31 satellites orbiting around the Earth in six equally spaced planes with an altitude of 20200 Km approximately. This system is developed, maintained, and controlled by the US space force. It is committed to assure the availability of 24 satellites 95% of the time by US department of defense.

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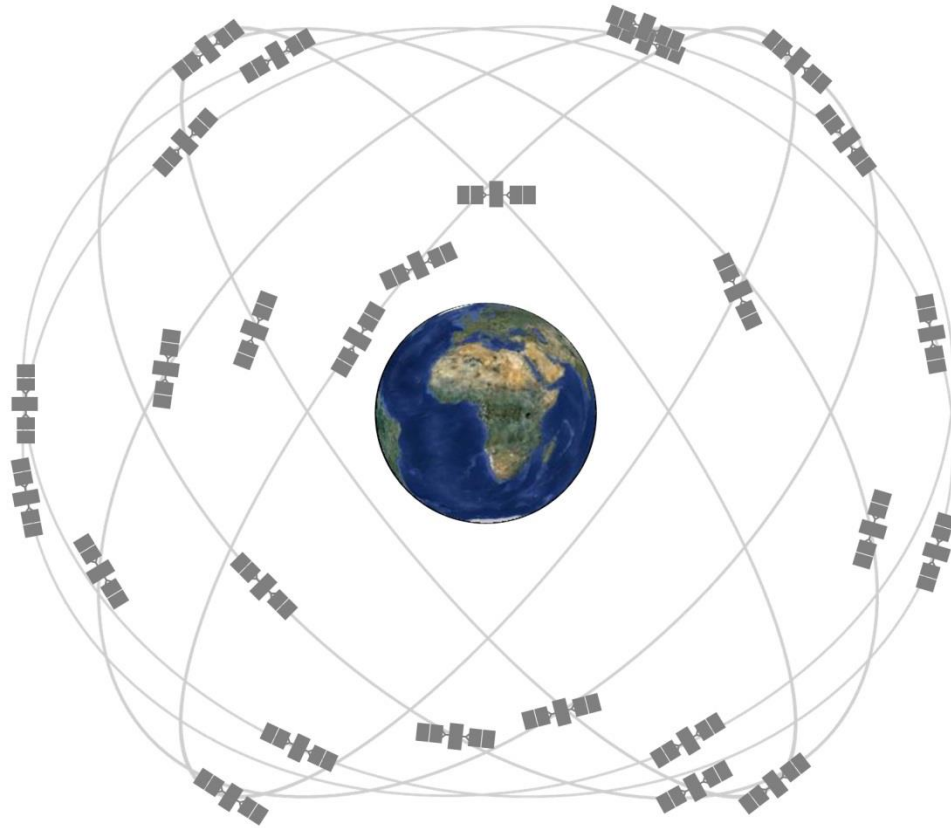


Fig 2.1: GPS Constellation in Space

2.1.1 Space Segment

The constellation is providing services using its both old and new satellites generations.

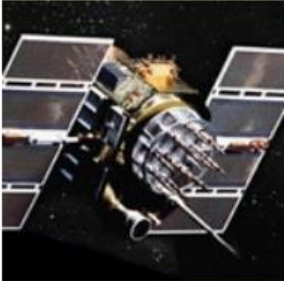
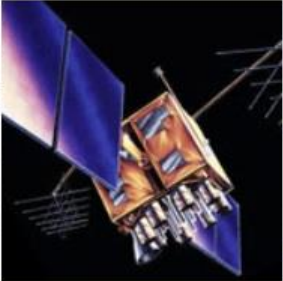
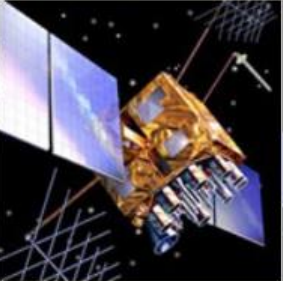
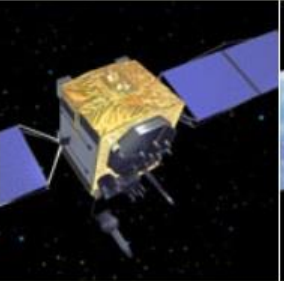
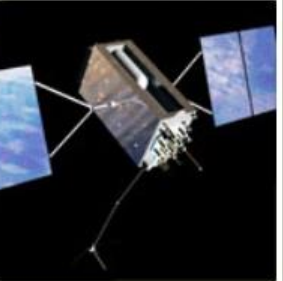
LEGACY SATELLITES			MODERNIZED SATELLITES	
				
BLOCK IIA	BLOCK IIR	BLOCK IIR-M	BLOCK IIF	GPS III/IIIF
0 operational	8 operational	7 operational	12 operational	4 operational

Fig 2.2: GPS operational satellites status

1. Block IIA (Advanced)

This is the 2nd Generation and advanced, Now 0 operational satellites. They had coarse acquisition code on L1 frequency for civil use and P(Y) code on L1 & L2 frequency for military use. They had 7.5 years of design life and launched in 1990-1997. The last one was decommissioned in 2019.

2. Block IIR (Replenishment)

The satellites belong to this block have C/A code on L1 frequency and military code P(Y) on both L1 & L2. They have onboard clock monitoring with 7.5 years life, and was launched from 1997-2004

3. Block IIR-M (Modernized)

The satellite belong this block has all the legacy signals, for better jam resistance, different power levels for military signals with 7.5 Year design life and was put into space during 2005-2009

4. Block IIR-F (Follow-on)

This constellation has similar signals as Block IIR-Modernized, with 3rd civil signal on L5 frequency, latest atomic clocks, better accuracy and signal strength, and quality, They have 12 years design life and was launched between 2010 and 2016

5. GPS III/III-F (Follow-on)

This constellation has similar signals as Block IIR-Modernized, with 3rd civil signal on L5 frequency, latest atomic clocks, better accuracy and signal strength, and quality, They have 12 years design life and was launched between 2010 and 2016[3]

2.1.2 Control Segment.

This includes two master control stations, 11 Telemetry, tracking command(TTC) antennas, and 16 TTC stations that track monitor data transmission, perform data analysis, resend data and commands to the constellation.[4]

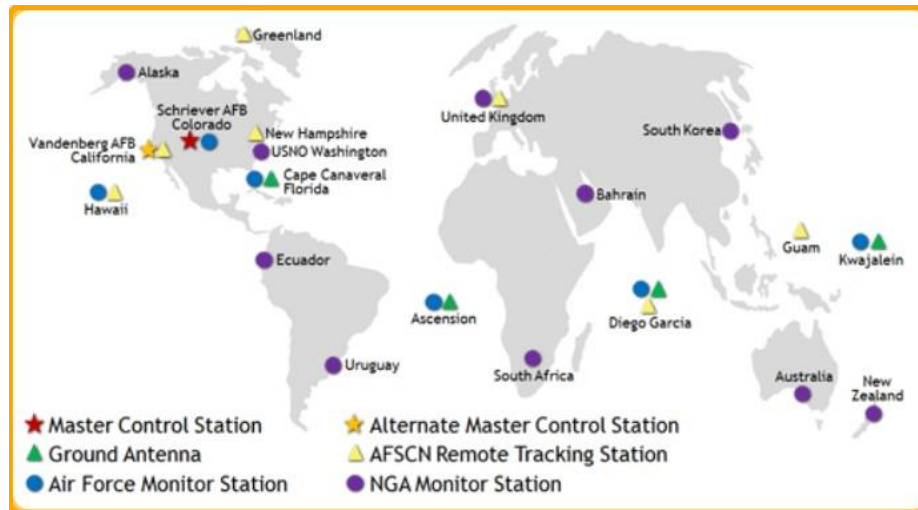


Fig 2.3 GPS control Stations map

2.1.3 GPS Signals

The purpose of modernization is the transmission of new signals for reliable and accurate positioning...

- 1) L2C is a new second GPS signal to fulfill the commercial needs of civilians. This signal is being transmitted by 23 satellites as of 15 June 2021. This can be easily received under trees and indoors because of its high transmitted power.
- 2) For the safety, transportation, and higher performance applications and aviation industry, as of 15 June 2021, 16 satellites from the constellation are broadcasting L5 signal.
- 3) To enable interoperability between Americans and other GNSS constellations. As of 15 June 2021, 4 satellites from the constellation are broadcasting the L1C signal [5].

2.2 GLONASS

GLONASS Stands for Globalnaya Navigazionnaya Sputnikovaya Sistema is a positioning, timing, and navigation utility developed, maintained, and control by the Russian Federation. This constellation consists of 27 satellites from which 23 are operational, 1 is under maintenance, 1 is spare and 2 are under flight test phase as of July 2021 [6].

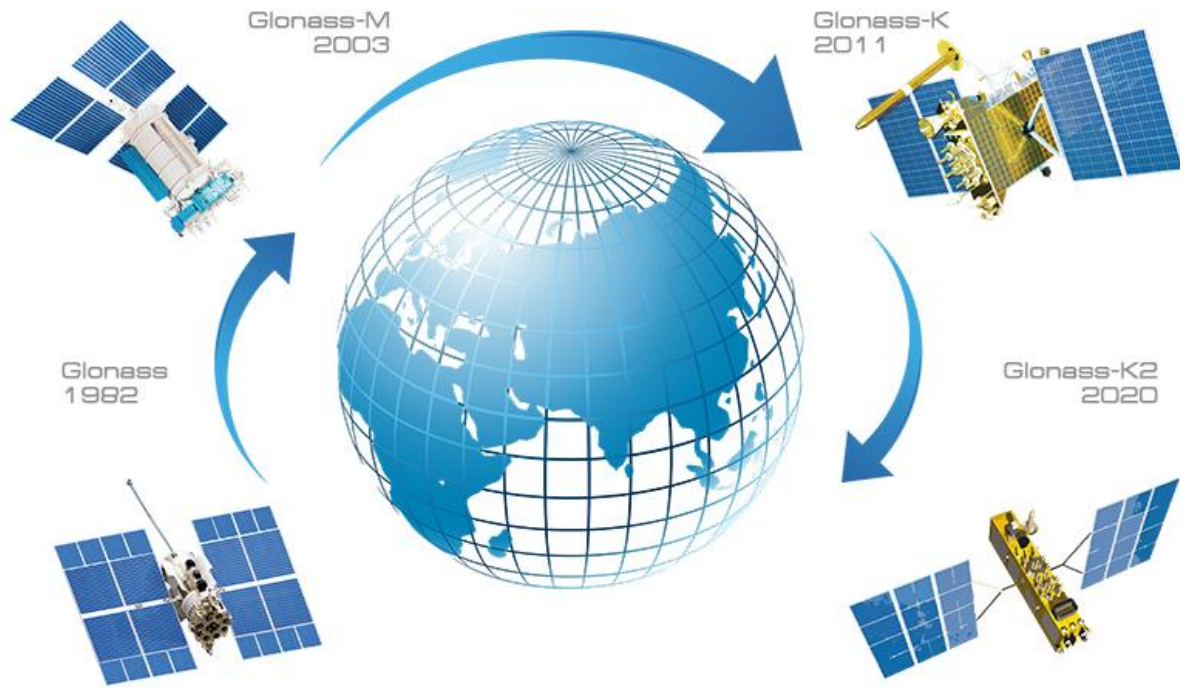


Fig 2.4: GLONASS Space System

Capabilities	GLONASS	GLONASS-M	GLONASS-K
Time of deployment	Between 1982-2005	Between 2003-2016	Between 2011-2018
Status	Decommissioned	In use	In Use
Orbital Parameters	Circular, Altitude 19,100km, Inclination 64.8 degree, Time period 11h 15min, 44 sec		
Number of orbital planes	24 satellites equally distributed in 3 different planes.		
Design life (Years)	3	7	16
Signal Type	FDMA	FDMA+CDMA for SVs 755-761	FDMA +CDMA
Frequencies for open Access signal	L1OF(1602MHz)	L1OF(1602MHz) L2OF(1246MHz) L3OC(1202MHz) for SVs 755+	L1OF(1602MHz) L2OF(1246MHz) L3OC(1202MHz) L2OC (1248MHz for SVs 17L+
Restricted Access signals	L1SF(1592MHz) L2SF(1237MHz)	L1SF(1592MHz) L2SF(1237MHz)	L1SF(1592MHz) L2SF(1237MHz) L2SC(1248MHz) for SVs 17L+

Table 2.1: GLONASS Characteristics

2.3 Galileo

2.3.1 Space segment

The constellation is consisting of 22 satellites with flying at the altitude of 23222 km and has time period of 14 hours and 56 degrees. It is developed, being operated, and maintained by the European Union (EU) for the provision of precise positioning, timing, and navigation solutions. Six to eight satellites are easily available all the time at any location of the Earth. The constellation is on track to completion in 2020 [7].

2.3.2 IOV & FOC satellites

The four satellites are in the orbit validation phase while the remaining 18 have been placed in orbits to reach full operational capability.

Characteristic parameter	IOV Satellite	FOC satellite
Mass(Kg)	About 700	About 733
Size with solar wings stowed (m)	3.02*1.56*1.59	2.91*1.7*1.4
Size with solar wings deployed (m)	2.74*14.5*1.59	2.5*1.2*1.1
Design life (year)	More than 12	More than 12
Available power (W)	1420(sunlight),1355(eclipse)	1900

Table 2.2: Characteristics of Galileo satellites

2.3.3 Control segment

The control segment contains 02 control stations suited in Germany and Italy. Each station consists of one ground segment for control functions and one ground mission segment for mission functions. The ground control segment performs spacecraft housekeeping and constellation maintenance with the help of a globally distributed network of TT&C stations while the ground mission and timing control monitor the navigation and timing data of navigation messages through Galileo sensor stations. The GMS communicates with the satellite by Galileo uplink stations[8].

2.4 BeiDou

The constellation has been developed and being operated by the Chinese Government. It is consisting of, ground segment, space segment and user segment. The space segment have geostationary, inclined geosynchronous, and medium earth orbits satellites [9].

2.5 Quasi-zenith Satellite System/Michibiki

The QZSS is a regional constellation in Japan consisting of four satellites in the Quazi-zenith orbit. However, the Quazi-zenith term is used for both Quazi-zenith orbit and geostationary orbit. Sometimes, it is also called the Japanese GPS. This four satellite

constellation has been in operation since November 2018. Minimum 3 satellites are always visible in Asia pacific region close to Japan and compatible with GPS. On 11 September 2010 first satellite was put into space and operated by the Japan Aerospace Agency (JAXA). Two more satellites with Quazi-zenith and one with geostationary orbit were launched in February 2017. In this way, four satellites constellation is functional since November 2018. It is also forecasted that this number will reach 7 in 2024 to approach precise positional accuracy [12]. QZSS only uses satellites that are near to zenith to improve accuracy in urban areas due to high buildings.

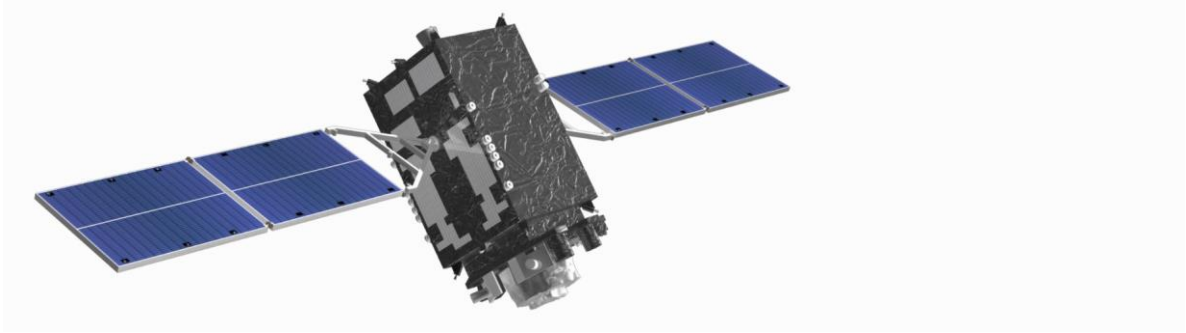


Fig 2.5: QZSS Satellite



Fig 2.6 Quazi-Zenith orbit for QZSS

2.5.1 GNSS View

It is a software released by the QZSS management in 2014 for the reproduction of satellite positions in the constellation at a given location and time. It displays the current location and time by default but the set time can be modified. Beside the QZSS, this can

display other all GNSS constellations and space-based augmentation system. In addition to the web version, it is also developed for the Android and iOS platforms [13].

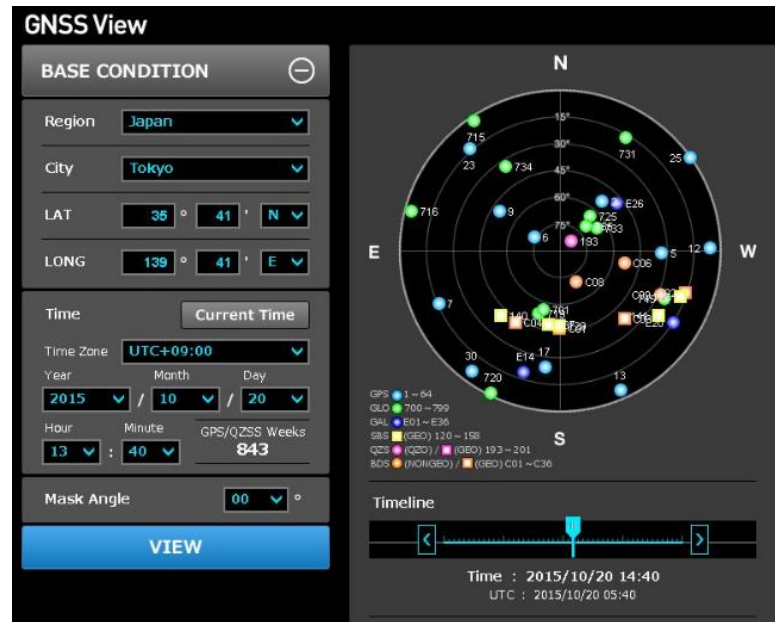


Fig 2.7 GNSS View User Interface

2.6 Indian Regional Navigation Satellite System (IRNSS)/Navigation Indian Constellation (NavIC)

It is a regional navigation constellation consisting of seven satellites, developed, operated, and maintained by India. It is designed to provide precise positioning services to India and primary service areas that are extended to 1500 km from its boundary. The IRNSS is expected to provide 20 m or better positional accuracy in its primary service area [14]. Three satellites are located in geostationary orbits in suitable slots and the remaining 04 are located in geosynchronous orbits into two different planes. In 2018, the IRNSS was declared to be fully operational. However, in 2016, India changed its name from IRNSS to NavIC (Navigation with Indian constellation [15]).

2.7 GNSS Augmentation Systems

Any external system which aids for the improvement in accuracy, reliability, or enhance positioning, navigation, and timing services. The Augmentation system is normally not inherently part of the GNSS system. Following are the well-known Augmentation systems.[16][17]

Augmentation System	Authority
Wide Area Augmentation System (WAAS)	US Federation Aviation Authority
Continuously Operating Reference Stations (CORS)	NOAA
Global Differential GPS (GDGPS)	NASA Jet propulsion Laboratory
International GNSS Service (IGS)	200 different organizations
European Geostationary Navigation Overlay Service(EGNOS	Europe
Multifunctional satellite Augmentation system(MSAS)	Japan
GPS Aided GEO augmentation navigation(GAGAN)	India
System for Differential Correction & Monitoring (SDCM	Russia

Table 2.3: Augmentation systems

3 GNSS RECEIVERS

The GNSS satellites placed in space transmit the signal using a polarized helical antenna having a right-handed circular configuration. The GNSS receiver on the ground or in space receives this weak signal, and then after amplification, it down-converts the IF intermediate frequency. Then the analogue signal is digitized with the help analog-to-digital converter (ADC). The antenna, low noise amplifier RF front end, and ADC are the hardware parts of a GNSS receiver. After the digital conversion, the baseband processing is performed using a software-based algorithm. At first, the code delay and carrier Doppler are determined in the acquisition phase. While the next phase includes the tracking of acquired satellites. Then the navigation data phase transitions the sub frames and navigational data can be downloaded that can be used for ephemeris and pseudo-range calculation in a conventional receiver both tracking and acquisition functions are performed in the hardware. Finally, the target position is calculated from the satellite position and pseudo-range.[20]

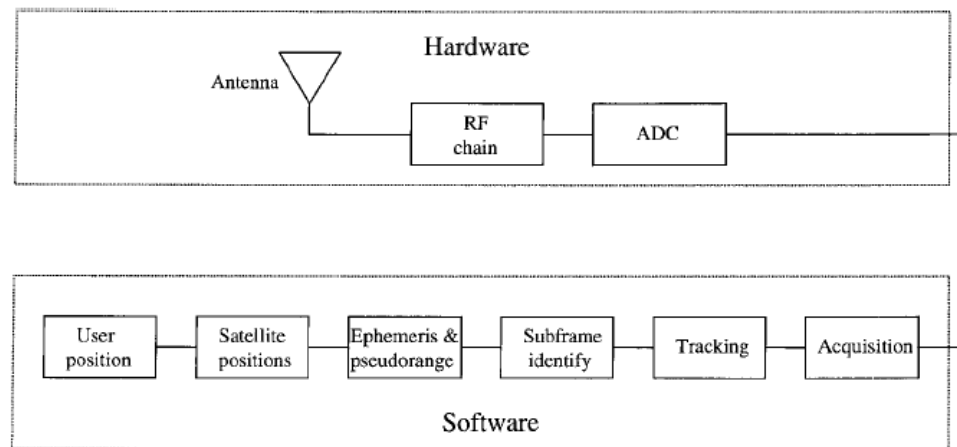


Fig 3.1 Basic GNSS Receiver

3.1 Antennas for GNSS

1. The GNSS broadband signal spread densely across the 1146-1616 MHz range. The commercial GNSS antenna varies available in the market have price range of \$4 to \$8000 and it might be higher than the cost of the receivers. The following are the some parameters of GNSS antennas for better performance in terms of radiofrequency.
2. **Operating Frequencies and bandwidth**
GNSS signals operating frequencies are ranged from 1146-1616 MHz with 470 MHz bandwidth, The GNSS antennas should be able to cover more Frequency bands.
3. **Gain pattern and polarization**

Fig 3.2 shows two elevation masks patterns, for low and high elevation angle. The GNSS antenna is being designed with higher cut-off elevation angles θ_c due to following reasons.

- I. Higher interference or noise near horizon
- II. More satellites are available near the zenith
- III. Super performance of receivers near the zenith

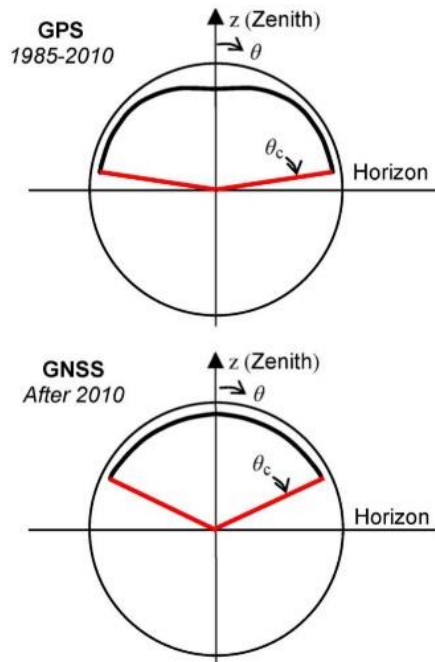


Fig 3.2: Antenna gain patterns for GPS antennas (1985–2010) and GNSS antennas (after 2010)

4. Multipath mitigation and interference suppression

The antennas are spatial filters which suppress multipath and interference signals to increase the S/N ratio of a line of sight signals below cut of angle θ . The choke ring, resistive load, and Good conductor ground plate are put together with high-performance GNSS antenna.

5. Phase center stability

The ultimate parameter that limits the performance of GNSS antennas is phase center stability this is especially important factor in high precision GNSS applications. Antenna phase center is the location or point on antenna where all the waves converge and converted to voltage accordingly. [18].

3.2 Basic Helical Antenna

For design a simple helical antenna the basic concept was introduced by Kraus in 1947. It has 3-Dimensional geometry and can be left or right-handed circle.

Parameter	Description	Relationship
D	Diameter of the helix	
C	Circumference of the helix	πD
S	Spacing between turns	
α	Pitch angle	$\tan^{-1} S/\pi D$
λ	One GPS Wavelength (190 mm)	
L	Length of one turn	
N	Number of turns	
A	Axial length	NS
d	Diameter of helix wire	

Table 3.1 Basic parameters of helical antenna

3.2.1 Right Handed Circular polarized (RHCP) Antenna

The antenna, low noise amplifier, and intermediate frequency conversion elements are in front of GNSS receivers. The antenna converts the incoming electromagnetic waves into electric currents that can be sensed through RF sections of GNSS receivers. Several antenna designs are possible for transmission and receiving of GNSS signals but signals transmitted from satellite have a low power density due to propagation through a long atmosphere from the height of 23000 Km approximately. An ideal RHCP antenna neither sends nor receives any signal transmitted by LHCP and vice versa...

3.2.2 Left Handed Circular Polarized antenna (LHCP)

GNSS satellites transmit the radio signals using direct sequence spread techniques. In case of GPS the signals are transmitted using right-handed circular polarized antenna on two carrier frequencies i.e., L1@ 1.57GHz and L2@1.23GHz approximately, using an array of 10 monofiler axial helical antenna S with the gain of 50W towards ground receiver. The power level is reduced to -160dBW when it is received and SNR is below 0dbW (-16dB). After reflection from any surface, the signal becomes further weaker and its polarization is reversed. The SNR of the reflected signal is the function of dielectric properties of the reflecting surface. If distance between observation and reflecting point

is taken as 25m, then the reflected signal will be 48dB below from a direct signal. The relationship between the range and the signal power can be expressed as below [19]

$$\text{Power (dB)} = 20\log_{10}(R/0.1)$$

[19] Has designed an LHCP antenna for GNSS reflected signals. Improving the antenna gain enhances the SNR of the signal. The weak signal generated by the antenna cannot be amplified with an inline amplifier because components amplify both noise and signal simultaneously and if the signal is weaker comparatively than after amplification, the signal will be suppressed by noise and cannot be detected. However, the amplifier can compensate for the losses due to the antenna cable and amplifier itself.

The important parameters for GPS LHCP antenna used for reflected signals have been discussed as a single functional element in their diagrams but no one has discussed its design, characteristics, and assembly in open literature in detail... Only [19] has documented its design assembly, laboratory, and field testing results of LHCP antenna for the reception of these weak reflected signals. I also followed the method described in [19] to develop an LHCP antenna that was used in the experiments...

3.3 Low Noise Amplifier (LNA)

Normally for the reception of GNSS Signal, we use a low noise amplifier to reduce the noise but in our case, we don't use any Low noise amplifier before antenna due to simulated signal.

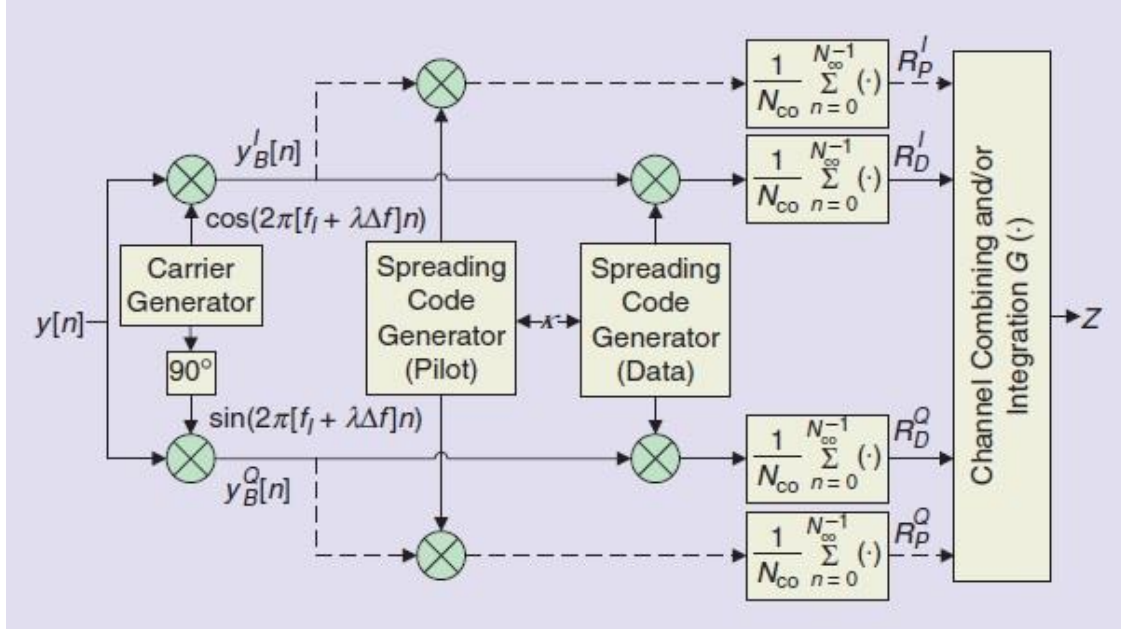
3.4 GNSS Signal Acquisition

Acquisition is the procedure to calculate the correct code delay and Doppler shift due to signal propagation and relative movement between satellite and receiver. The signal acquisition is performed by correlating the incoming and locally generated replicas.. The GNSS signal is declared to be detected when a strong correlation exists between the receiver and locally generated replicas and the correlation function peak exceeds from required threshold.. The parallel processing for both code delay and Doppler shift makes the receiver more fast and efficient [21].

Carrier to Noise (C/No) of GPS L1 code is about 45dB-Hz in an open sky line of sight observation while it is reduced to lower than 20dB-Hz for indoor environment of reflected signals. The acquisition function in a receiver searches for both code delay and Doppler and detects the incoming signals to track the satellites for position measurement.

Long spreading periodic codes modulates the Carrier signal w.r.t phase and time as well as Doppler frequency shift generates in the incoming signal due to satellite and receiver relative motion. The measured phase and Doppler should be within the fine resolution to track to initiate the tracking function successfully. The acquisition function tests every

supposed value of code delay Doppler within a search space using a correlator for the detection of incoming signals [22]



.Fig3.3: A diagram for the correlation, combining, and integration function to test a hypothesis. The Superscripts I and Q represent the in-Phase and Quadrature phase components

In case of reflected signals and indoor environment the acquisition procedure can be both delayed and costly. In normally the code phase hypothesis, the change step size Delta T is set half code chip. On the other hand, the number of Doppler frequency hypotheses can be more due to large relative motion and Doppler–frequency search step Size Delta f is small where Delta f is inversely proportional to the coherent correlation interval TCO...

$$y(t) = \begin{cases} AD(t-\tau)P(t-\tau)\cos(2\pi(f_I+f_D)t+\theta)+v(t), \\ A_D D(t-\tau)P_D(t-\tau)S_D(t-\tau)SC_D(t-\tau)\cos(2\pi(f_I+f_D)t+\theta) \\ +A_P P_P(t-\tau)S_P(t-\tau)SC_P(t-\tau)\sin(2\pi(f_I+f_D)t+\theta)+v(t), \\ [A_D D(t-\tau)P_D(t-\tau)SC_D(t-\tau)+A_P P_P(t-\tau)SC_P(t-\tau)] \\ \times \cos(2\pi(f_I+f_D)t+\theta)+v(t), \end{cases}$$

In the case of GNSS, reflected signals or in-door environment longer TCO is necessary, while the number of Doppler-frequency hypotheses increases consequently. As a result,

Acquisition time and complexity increase exponentially for reflected signals.

3.4.1 Serial Time Domain and Frequency-domain parallel Search

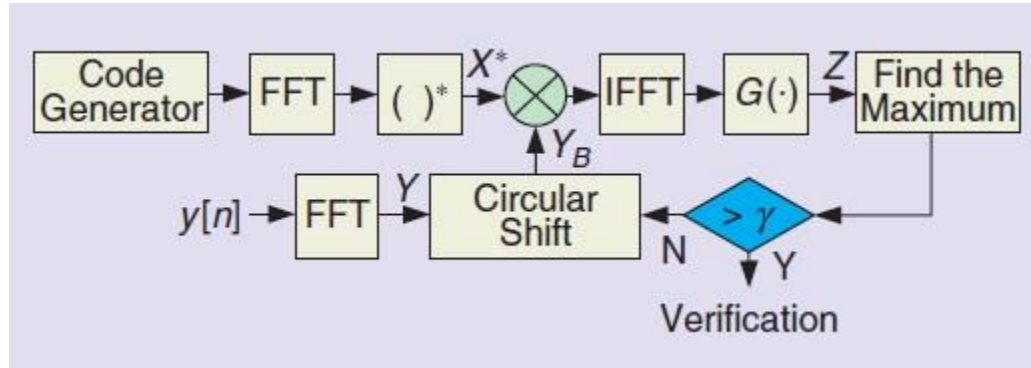


Fig 3.4: Frequency domain Parallel search domain

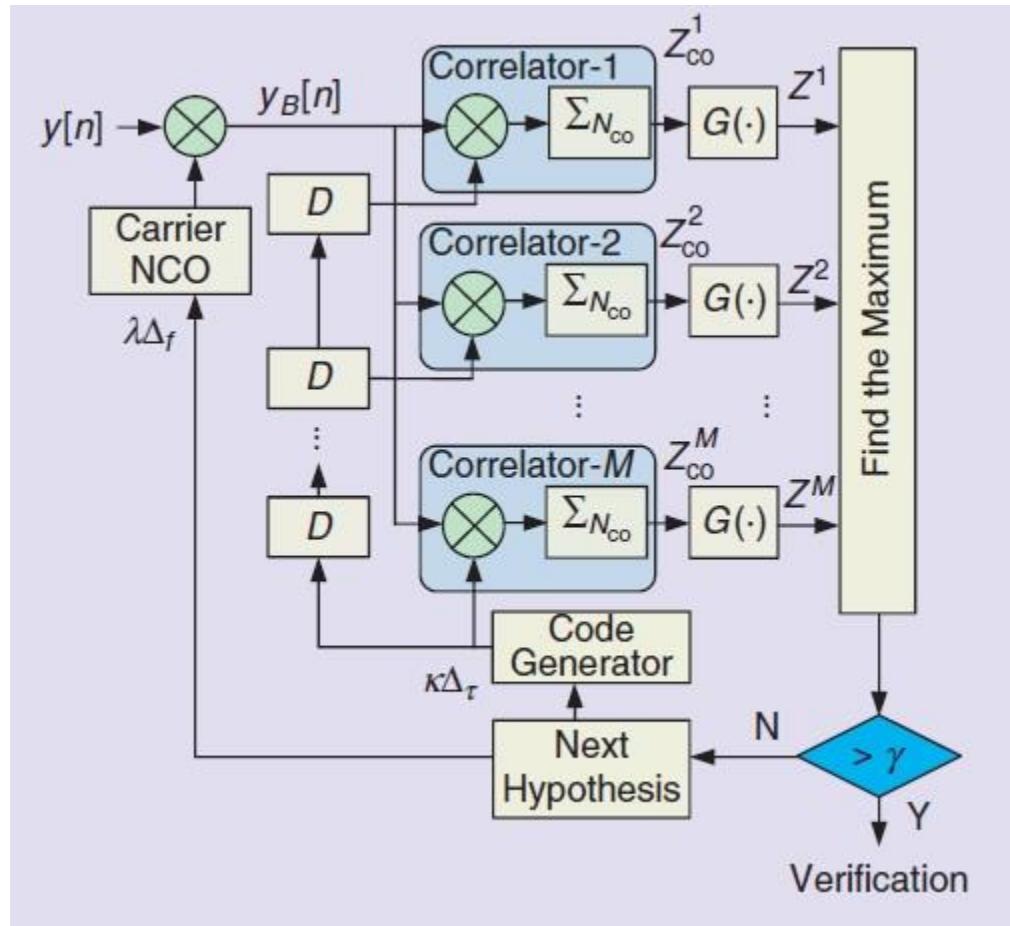


Fig 3.5: Time domain Serial search domain

4 GNSS ALTIMETRY

The Global Navigation satellite system was created 30 years ago for precise positioning, timing, and navigation purposes only but later on the GNSS system is used in a large variety of earth applications remote sensing. Similar to traditional satellite remote sensing the GNSS reflected and refracted signals can be used for monitoring of different land covers and earth atmospheres such as Snow, Ocean, Vegetation land, ice ionosphere, and troposphere. The GNSS L band Frequencies lie in the Microwave region which can penetrate from clouds. In 1980,s first time GNSS signals were suggested for radio occultation

In 1998 Martin-Naira was the first who proposed GNSS altimetry application. Garrison and Katzberg sensed ocean surface roughness and relating winds using GNSS reflected signals from the aircraft experiment. [23]. GNSS reflectometry and GNSS Radio Occultation are two different applications of remote sensing. The GNSS-R detail will be presented in the next paragraph.

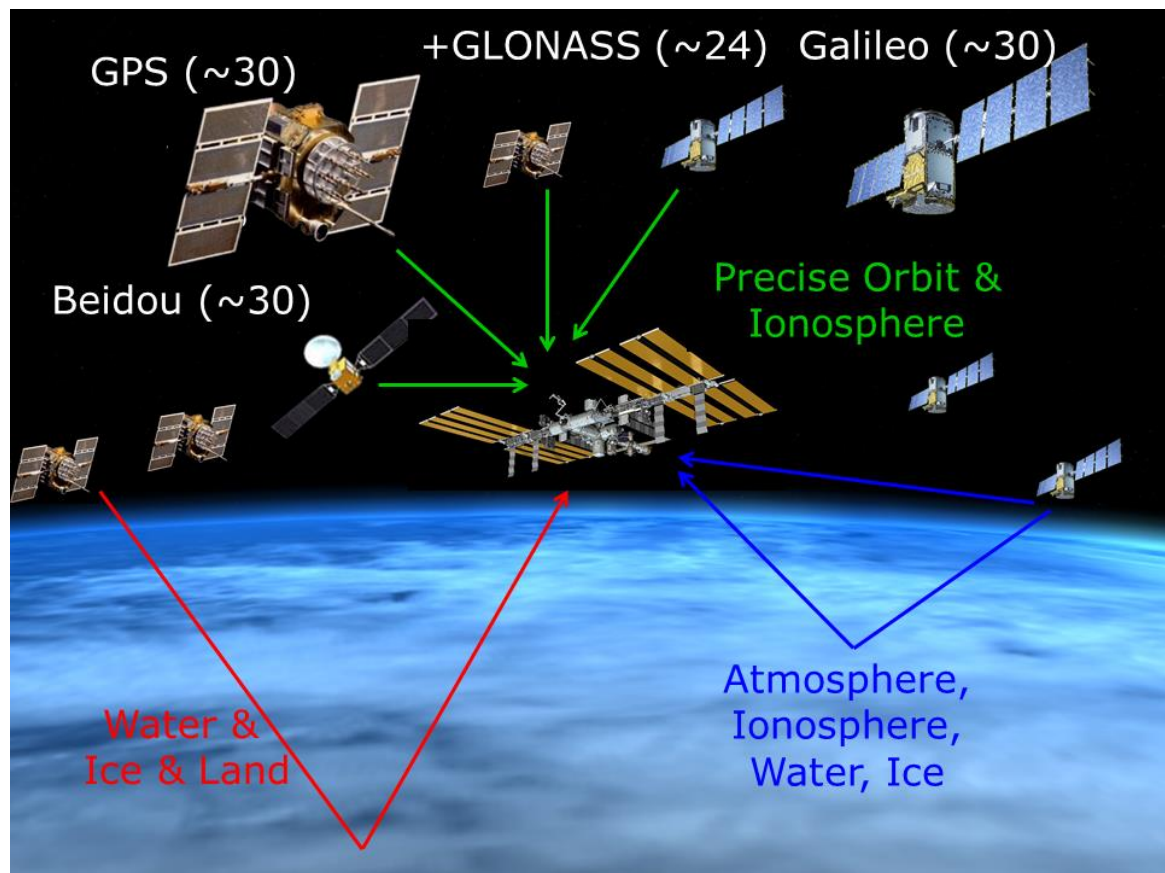


Fig 4.1: Different applications of GNSS remote sensing

4.1 GNSS Reflectometry

Generally, a multipath signal in GNSS positioning is major source of error. That undesirable phenomenon needs to be suppressed. GNSS reflected signal is one kind of multipath that is used for different Remote sensing applications. It is the method of microwave remote sensing that uses an L-band signal reflected off from various types of land covers to extract useful information.

The GNSS Reflectometry working principle is similar to Bistatic Radar. Most traditional Radars used for monitoring air space, and weather forecasting are monostatic. as transmitter and receiver are located on the single platform. On the other hand, the Bistatic radar system uses separate platform for both transmitter and receiver. The power of the receiver can be expressed through the Bistatic radar equation.[25]

$$P_{pq}^r = \frac{P_t \lambda^2}{(4\pi)^3} \int \frac{G_t G_r}{R^2 R_0^2} \sigma_{pq}^0 dA,$$

Where

P_t is Transmitted power

G_t is Transmitter Gain

G_r is called Receiver Antenna Gain

R =Range from transmitter to the target under observation

R_0 = Range from target to the receiver

σ_{pq}^0 = Polarization Dependent Biostatic radar coefficient

P =incidence polarization

q =Scattered polarization

The advantage of this technique over traditional remote sensing and radar altimetry are as follows

All GNSS satellites are used as a transmitter so an additional Transmitter is required

1. Plenty of signal sources including the 30 GPS, 30 Galileo, 24 GLONASS, and 30 Beidou are available globally in all weather and time.
2. Weak reflected signals can be acquired due to the Spread spectrum communication technique

4.2 Characteristics of GNSS-Reflected Signals

GNSS reflected Signals have a unique signature or set of characteristics that differ from direct signals.

Polarization:

The GNSS satellite transmits the signal at right-hand circular polarization but after reflecting from the land surface the polarization reversed and we used LHCP antenna to collect these weak signals.

Code delay and Doppler shift. The code delay introduced in reflected signal due to extra path segment prorogated and Doppler due to satellite movement and reflection.

Correlation: Correlation characteristics can be expressed by the integral function of correlation.

For direct signal

$$Y(t) = \int_0^{T_i} a(t) \cdot s(\tau) d\tau$$

But for the reflected signal, it is:

$$Y(\tau, \tau_0, t) = \int_0^{T_i} a(t + \tau_0 - \tau) \cdot s(t + \tau_0) dt$$

where

$a(t)$: local C/A code sequence

$s(t)$: data received by the receiver

T_i : time interval of integration

τ_0 : estimated code delay of specular point

τ : code delay within $\{-M, +N\}$ which need to be determine by the user.

The GNSS applications can be classified into four types [26]

1. Classification w.r.t observed surface i.e. ocean, land, ice, and snow
2. Classification w.r.t final product i.e. altimetry, roughness, permittivity parameters such as humidity, and temperature, permittivity, defines the reflectivity levels of the surface material
3. Observables used in the techniques also characterize the classification i.e. power amplitude and phase
4. It can also be classified w.r.t Polarimetric and Non-Polarimetric observables

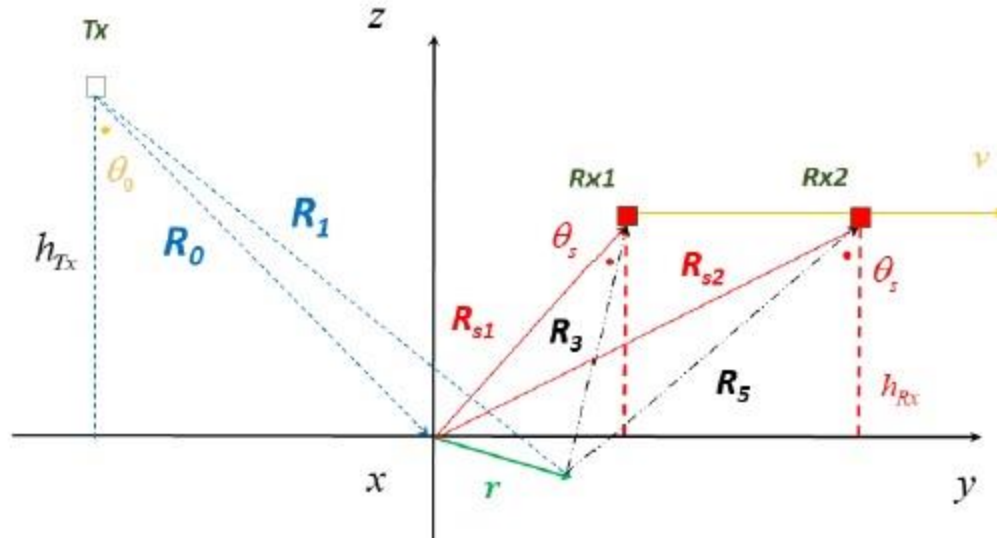


Fig4.2: Scattering model for GNSS-R system.

In the case of Near Specular Scattering, the GNSS reflected signals from different land covers are partially coherent and essentially incoherent in most cases. The correlation time depends upon roughness parameters, namely standard deviation, and correlation length. Losses of coherence influence system performance and data exploitation. Like other systems, the performance of GNSS-R can be significantly affected by the decorrelation of signals. Typically by considering both transmitter and receiver steady and reflection from a random roughness surface. The received reflected signals can be decomposed into Coherent and incoherent signals. The first one is originated by the listing zone while the coherent signals are generated by the plane surface.[27]

Apart from Remote sensing and SAR applications, the detection of moving targets is also an important application this has been already experimentally demonstrated using airplanes. Traditionally passive coherent location radars used for airborne and maritime target detection all are based on terrestrial signal sources .these sources provide persistent monitoring of their surrounding areas All these sources have a common drawback that they cannot provide coverage in the open sea. The figure demonstrates the concept of GNSS-based radar for maritime target detection.[28]

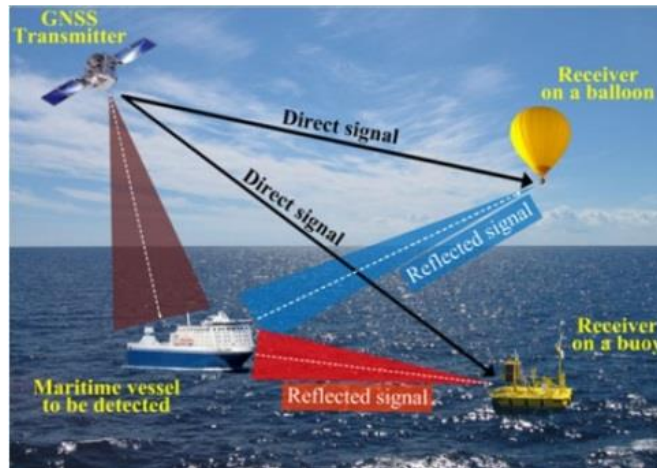


Fig 4.3 concept of GNSS-based radar for maritime target detection

4.3 Altimeter

It can be defined as the elevation measurement system used by mountaineers, skydivers, and flying machines. Its simple purpose is to describe that what the height of the device or person from sea level. There are two Basics types of altimeter, Barometric and Radar altimeter.

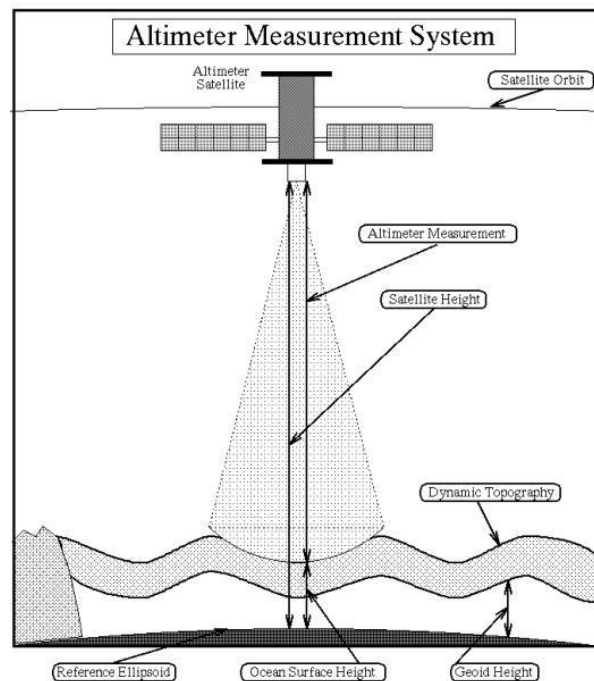


Fig4.4: Altimeter Measurement System

1. The Barometric altimeter uses the mercury meter level which changed with the change in atmospheric pressure. As the atmospheric pressure depends upon

Atmospheric height. The mercury level increases 1 mili bar with the increase of 8.23 m height in the atmosphere

2. The Radar altimeter uses the reflected microwave's travel time from target to receive. As the waves travel with the speed of light so by knowing the pulse return time we can measure the distance between target and platform.

The barometric altimeters are mostly used for low altitude but higher altitudes Radar-based is preferred due to greater reliability required [29]

4.4 Radio Altimeter

Radio altimeter is used as an auxiliary of the barometer in commercial and manned airplanes and aids in landing and approach. In radio altimeters, the altitude measurement is made by frequency-modulated continuous-wave radar. Normally airborne radio altimeters are designed at low power, low range FM-CW radar and operate in 4.3 GHz.Microwave band. The resolution and accuracy are determined by signal bandwidth. An FM-CW radar uses saw tooth or symmetrical triangle wave as a modulating signal [30]

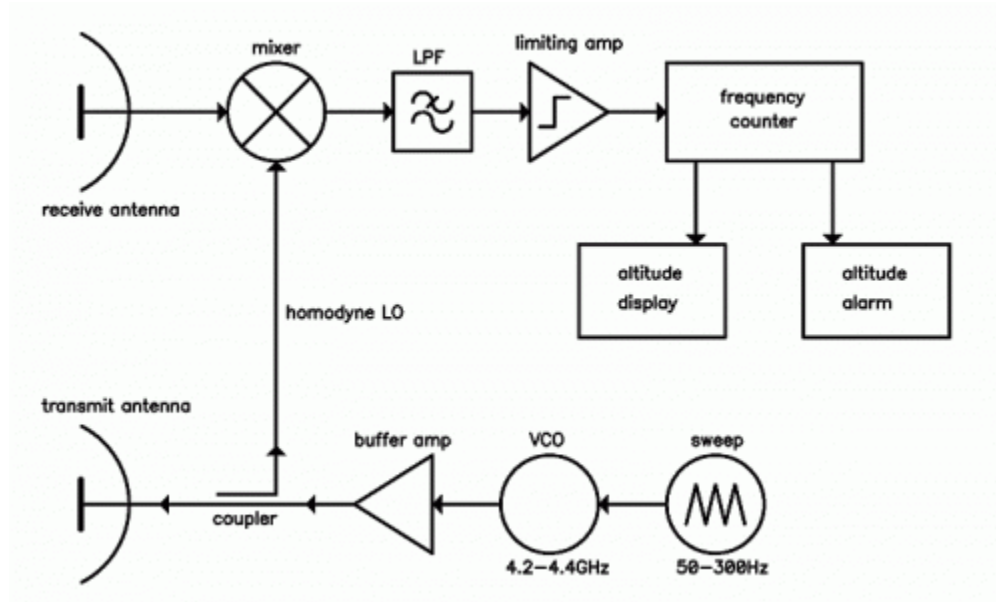


Fig 4.5: Conventional FM radio-altimeter design

4.5 GNSS Altimetry

The GNSS Altimetry technique's performance depends upon the signal-to-noise ratio of reelected signals. Therefore these can be only applied for strong reelecting surfaces such as water smooth ice and land at low altitudes. The GNSS-based altimeter measures the vertical distance between the receiver and the surface or the vertical location of the specular point w.r.t a certain reference system. In the space & time domain, the

observables for altimetry are called ranges and delays, and both are related to each other by the speed of light. The altimetry range is the distance traveled by the reflected signal w.r.t distance traveled by direct signal. The General form of the altimetry equation is as

$$\text{follows } R_{r-d} = p_{\text{geo}} + p_{\text{atm}} + p_{\text{ins}} + n$$

Where

R_{r-d} = measured altimetry range

p_{geo} = Geometric distance between direct and reflected ray paths

p_{atm} = Sum of delays introduced by troposphere and ionosphere

p_{ins} = Instrumental biases such as clock offset cable length and antenna offset vector

n = noise

The simultaneous observations from different satellites correspond to different azimuths and angles of observations. Both Ionospheric and tropospheric effects can be neglected since both direct and reflected signals are similarly affected. The GNSS Based multistate radar can produce 100 simultaneous measurements as compare to traditional altimeters which can only produce one or two. Even with large precision errors, the accuracy can be achieved through averaging improving the height precision.[31]

4.5.1 Group delay Altimetry observables

When the range is measured in terms of delay through code correlation functions it is called group delay or pseudo-range. This observable is only useful for smooth surface reflections event and is difficult to apply for rough surfaces.[32] Used BDS-3 new civil signal for GNSS altimetry and showed that the accuracy can be achieved about 10-20cm. The data was collected using a front end with a 62MHz frequency. This new signal has wider bandwidth. The precision was evaluated using a 26GHz radar altimeter

4.5.2 Phase delay Altimetry observables

The phase delay altimetry technique is used to measure the range through carrier phase observations at the antenna phase center. The range is measured from phase difference

for both direct and reflected signals. From rough surfaces the signal phase becomes non-coherent and cannot be applied.[26]

[33] Have demonstrated the concept of carrier phase altimetry by developing open-source software. Carrier phase-based complex correlator output represented by plane-phasor diagram constructed with the help of signal processing. The direct signal is located on the x-axis positive side. In the ideal case reflected power is constant and it should be located on the circle centered on the origin. The angle between the I-axis and reflected signal gives the reflected minus direct signal CPM

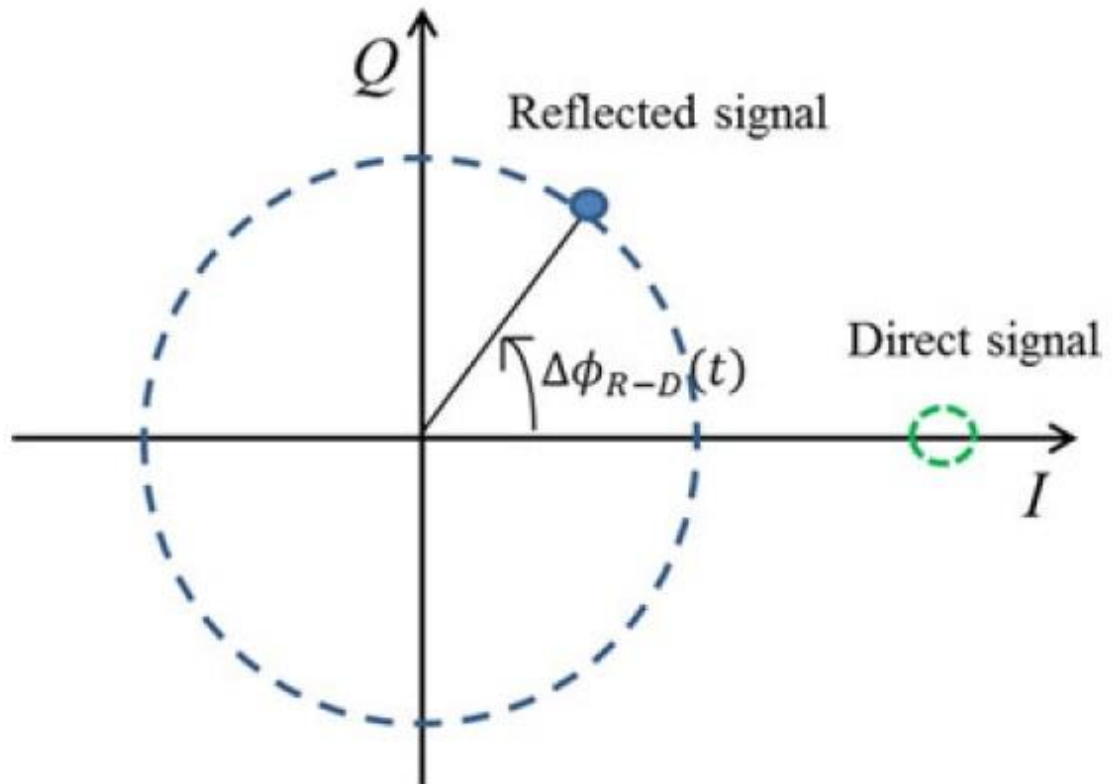


Fig4.6: Carrier phase-based complex correlator output

GNSS based altimetry have several advantages over mono-static active radar systems i.e. reduced cost, multiple experiments/samples on large area, highly accurate space-time reference frame. On the other hand, it has also some limitations i.e. The GNSS signals are not optimized for remote sensing applications resulted in limited bandwidth, lower signal power, and therefore lower altimetry accuracy and precision. Carrier-based altimetry produced precise and accurate results while code-based measurement resulted in either imprecise or biased data.[34].

The GNSS-R code delay-based altimetry is limited by received Low power signals and the small bandwidth of the reflected signals. The power of the reflected signal is much lower than the direct signal and thus down looking antenna with high gain is necessarily

required. In addition, the bandwidth is much narrow i.e. 2-20MHz vs 300MHz of Traditional Radar altimeters implying poorer one-shot altimetry precision.[35]

Conventional satellite altimeters measure the distance from platform to ground surface through round-trip time of radar pulse along the nadir line. These altimeters use a nadir-looking configuration, larger bandwidth, and high microwave carrier frequency due to being less affected by the ionosphere. On the other side, GNSS-based radar altimeters use an off-nadir configuration, low transmitted power, narrow bandwidth typically bands specified for radio navigation. These limitations result in poorer height resolution for GNSS-based radar altimeters (20-30cm). Although the Spatio-temporal sampling can be increased by receiving reflected signals from different incident angles.[36]

4.5.3 SNR-Based GNSS Reflectometry

[37] Has demonstrated the concept of GNSS altimetry based on Signal to Noise ratio (SNR) using the observations taken from ground-based GNSS stations for altimetry applications. They have also demonstrated the complete hardware and software development for SNR based altimetry to provide experimental validation

5 DATA ACQUISITION AND METHODOLOGY

For GNSS Altimetry the actual data has to be acquired and saved in a digital format. Later on, it is manipulated during signal processing. The hardware consisted of a high gain and directional antenna for acquiring the reflected signals and an RF Front end has been used for capturing the data at suitable IF and sampling frequency. This chapter will contain some details regarding the hardware and Data acquisition aspect of the altimetry system.

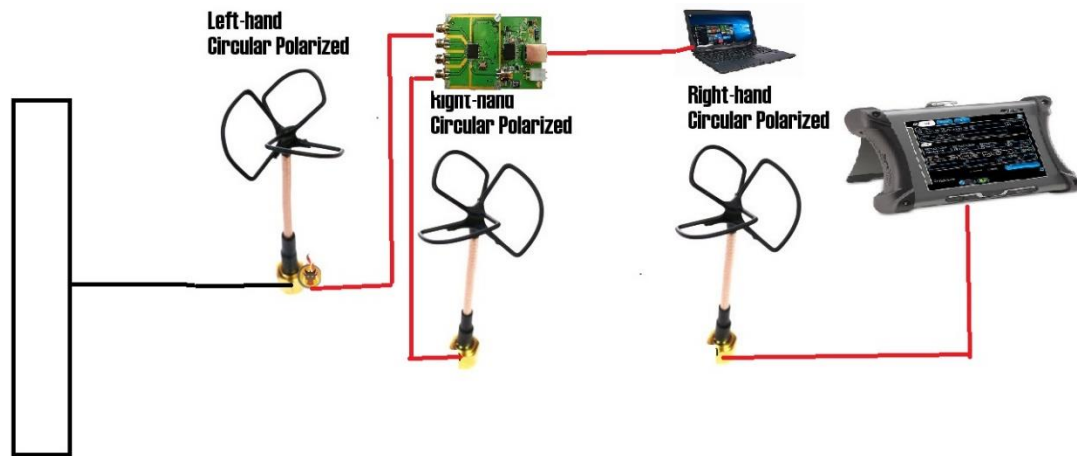


Fig 5.1 Schematic diagram of Altimetry experimental setup

The altimetry setup can be divided into 5 basic components. The GPS Simulator for generating the required signals, RHCP Antenna for signal transmission, LHCP antenna for receiving the reflected signal and RF front end for acquiring the signal in digital format, and finally the software algorithms for signal processing.

The GPS Simulator transmits the signal by using an RHCP antenna after bouncing back from the reflecting surface the amplitude is reduced and since the reflection coefficient is less than 1 and some of the signals are absorbed. The Equipment used can be divided into 5 Basic components which we will explain here:

5.1 The GNSS Simulator

The GNSS simulator was used to simulate the required signals as the experiments were performed inside a lab. The GNSS Simulators can be categorized as single-channel and multi-channel Simulators.

5.1.1 Single channel simulator

This type of simulator is also known as a signal generator capable of simulating a signal from one satellite only. It usually controls a signal Doppler profile. It can also be called a satellite transmitter with the only difference of Doppler shift control. In a single channel,

the carrier signal is modulated by code and navigation messages. A single channel simulator can also be used as a Pseudolite.

5.1.2 Multi-channel Simulator

It can also know as a constellation simulator. It coherently simulates multiple satellites signals. This type of simulator is mostly used for R & D work and by manufacturers and designers

There are three key features defined by a multi-channel simulator

1. The capability of recreating GNSS RF signals to be indistinguishable from real signals
2. The capability of providing repeatability of signal generations
3. The ability of a user to control most of the simulation environment [38]

5.1.3 GPSG-1000

The Configurable simulator Capable of simulating GPS L1, L2C, L1C, L5 & Galileo E1, E5, E5a, E5b and even both can be simulated simultaneously. It is also capable of static & dynamic simulations Via NEMA-0183 and a capable file record for playback. It is also capable of reconfiguring satellite vehicle parameters with an unlimited number of plans and a built-in GPS Receiver that allows positional simulation in the form of range and heading.



Fig 5.2: Aeroflex® GPSG-1000 Simulator

5.2 Amoungo NUT4NT

It is an open-source 4 channels GNSS Receiver board developed by Amungo Navigation with the implementation of the world's first low-cost RF Front end solution for Global Navigation satellite system with 4 channels supporting all GNSS systems and frequencies... (Electronics-lab.com, 2021).

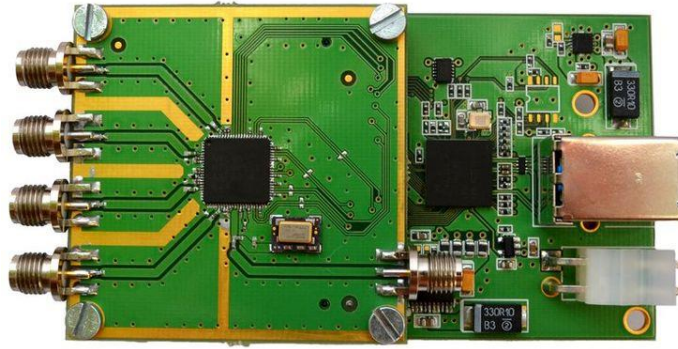


Fig 5.3: Amoungo NUT4NT Dual input mode (left), Four input mode (Right)

5.2.1 Hardware and Software Specifications

1. NT1065 Receiver Module
2. CYUSB3014 Controller
3. 10MHz Clock rate
4. 2 bit ADC Resolution
5. Up to 99MHz ADC Frequency
6. 20-100Mbytes/Sec Transfer rate
7. 5v@5A Power requirement
8. All GNSS Systems with all Bands
9. 4 Channel synchronous reception for antenna array processing.

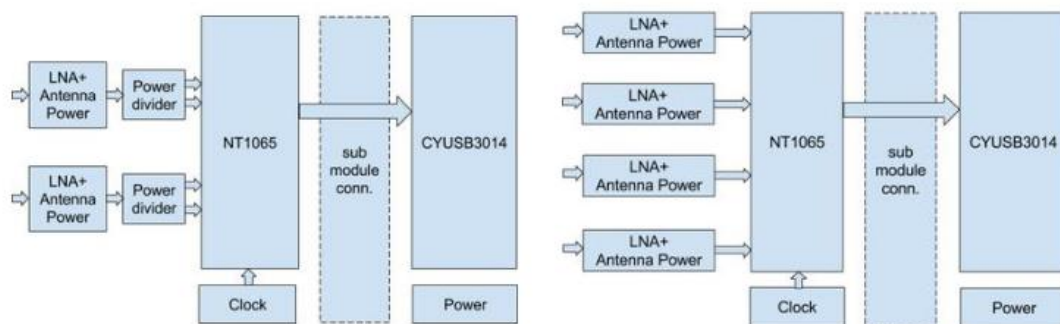


Fig 5.4: Amoungo NUT4NT 4 channel RF front end Board

5.2.2 NT1065 (GNSS RECEIVER DESIGNED BY NTLABS)

Typical SDR-based receivers use an integrated module for digital signal processing including amplification frequency conversion sampling and digitization using FPGAs, Microprocessors, and DSP or combination. But NT1065 is an Advanced Front end module that performs the necessary processing of GNSS signals within four independent channels which benefits for processing of multisystem GNSS signals.

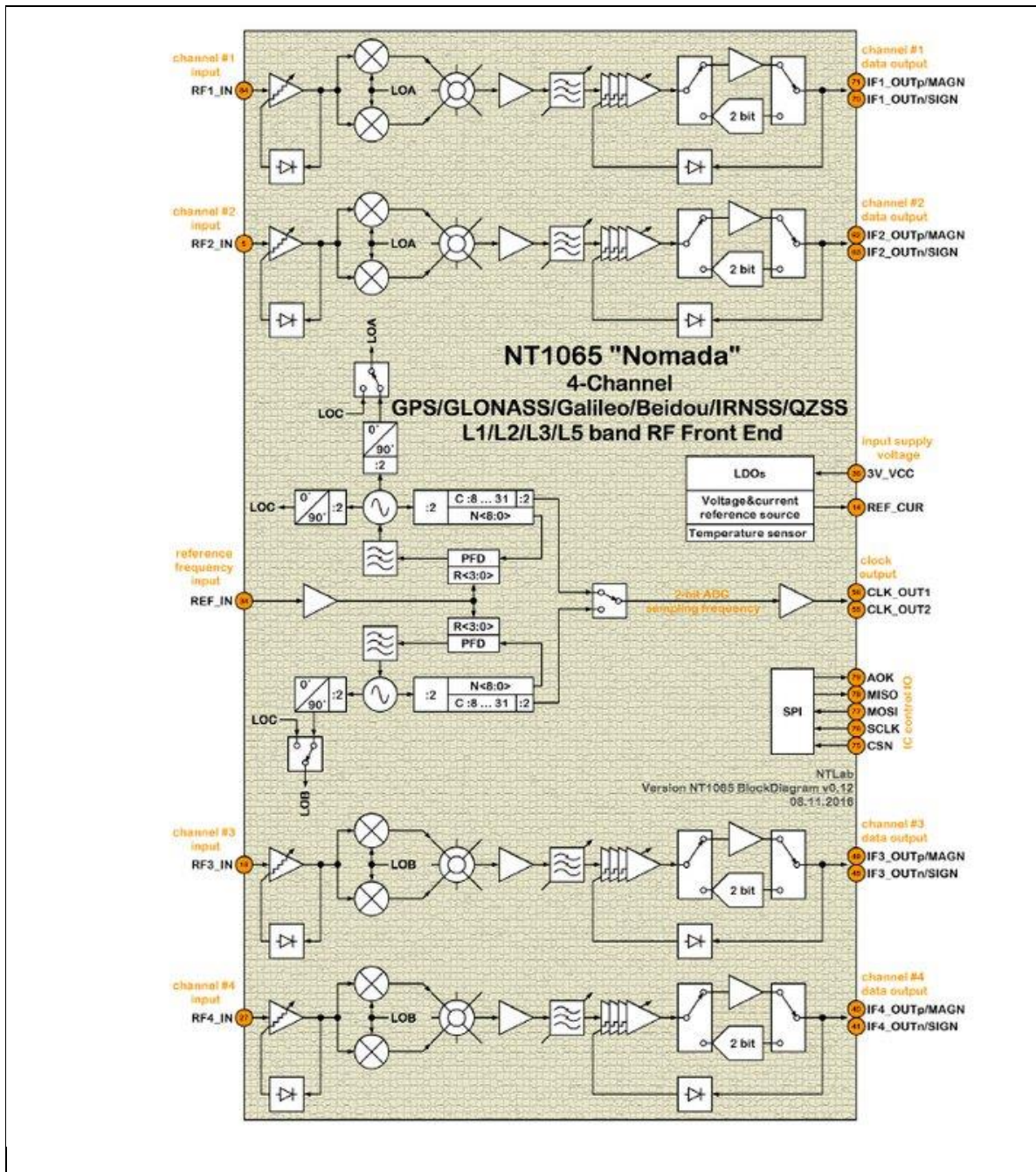


Fig 5.5 Block Diagram of NT1065

NT1065 is an efficient Front-end module for SDR-based GNSS Receivers. Results showed that NT1065 is an appropriate possible option.[39]

5.3 Right-Handed Circular Polarized (RHCP) Antenna

For the signal transmission from the simulator, RHCP helical antenna was used, similarly to the one used by GPS satellites. Since the deployment of all GNSS systems, GNSS applications have proliferated all over the world including the military as well as commercial and consumer market. So the Importance of a GPS antenna for the receiver is obvious [40]

The antenna collected the electric current sensible to RF and converts them into electromagnetic waves.

5.4 Left Hand Circular Polarized (LHCP) Antenna

The GPS satellite transmits Signals using DSSS (Direct sequence spread Spectrum) techniques on two carrier frequencies named L1 @ 1575.42MHz and L2 @ 1227.6MHz. The GPS satellite transmits the signal towards earth with a gain of 50 watts. Due to the long-distance traveled by signal, the power is reduced to -160dBW. The reflected signals are reduced in power and reverse their polarization after reflecting from the target and become LHCP [19].

As shown in the figure of the complete experimental setup a custom-made high gain LHCP helical antenna was used for acquisition of reflected Signal as the simulator transmitted RHCP signals are converted to LHCP after reflection from the wall.

For detection and acquisition of the weak reflected signal a custom LHCP helical antenna was designed with (9 dBi) of pure antenna gain. The PVC Pipe having an outer circumference equal to one of GPS wavelengths was supported with 2.0 mm enameled copper wire utilized as the helix coil without interference with the direct signal. A 2 mm wire turned out to be 0.01 Lambda. A 2 mm thick aluminum (190mm x 190mm) sheet having 1 Side equal to GPS wavelength was used as the base plate. The antenna wire is connected to an RG-58 Coaxial Cable via an SMA connector.

Description	Parameter	Value
Diameter of the helix	D	60mm
Circumference of helix	C	190mm
Spacing between turns	S	47mm
Pitch angle	A	140
Number of turns	N	9
Axial Length	A	1000mm
Diameter of helix wire	D	2mm
Gain w.r.t Isotropic		9dBi

Table 5.1: characteristics of LHCP helical antenna

5.4.1 Antenna Gain

The antenna Gain is directly proportional to the number of turns of the coil. There are multiple formulas for gain calculation in technical literature. As we used a Simulator so our signal is very strong as compared to the actual signal. For Actual signal antenna with 21 turns is used with the gain of 120db. [19].

$$G = 15NS\pi^2 \frac{D^3}{\lambda^3}$$

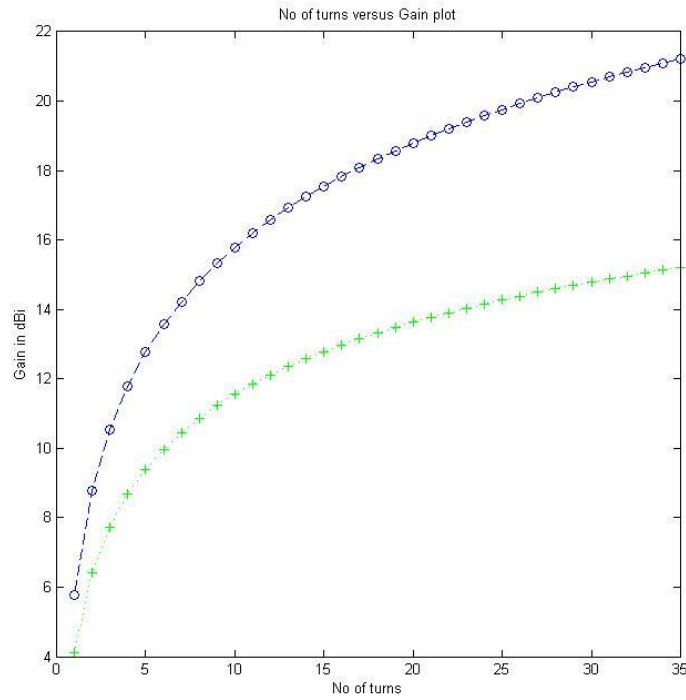


Fig 5.6: Antenna Gain against Number of turns

Another formula for the gain of the helical antenna in dBi is mentioned in [19] and reproduced below

$$Gain(dBi) = 11.8 + 10\log_{10}(C \lambda^2 N S \lambda)$$

By using Netherland design formula with 9 Number of turns the gain of our antenna is 8.9dBi

5.5 Chip Resolution

As the C/A code contains 1023 chips and is transmitted with frequency of 1.023MHz, and we know that

$$C = f\lambda$$

By using the speed of light as 3×10^8 m/s the length of a single chip is 300m. As the sampling frequency of Amoungo is 53 MHz, therefore the resolution of the GPS altimeter is 5.7m. By using the front end with a higher sampling frequency the resolution can be further increased. During the acquisition, multiple experiments were performed

5.6 The software algorithms.

The algorithm performs coherent acquisition on collected data of any length for direct and reflected GPS signals. First, we have to enter the Satellite number which we want to acquire. The acquisition for both direct and reflected signals is plotted and then compared in terms of chip delays. Frequency domain acquisition is performed as most of the latest receivers use it. The User has to enter the processing time for both signals. The algorithm Perform acquisition on collected data by dividing it into number of chunks so that it can be easily managed. Also for avoiding the OUT OF MEMORY message.

6 RESULTS AND ANALYSIS

This Chapter presents the results and analysis after data processing. The main objective of my research work is to collect and analyses GPS L1 Reflected signal for altimetry proposes. The next section of the chapter will present experimental results and analysis.

For ease of understanding, this research work can be divided into four (04) steps.

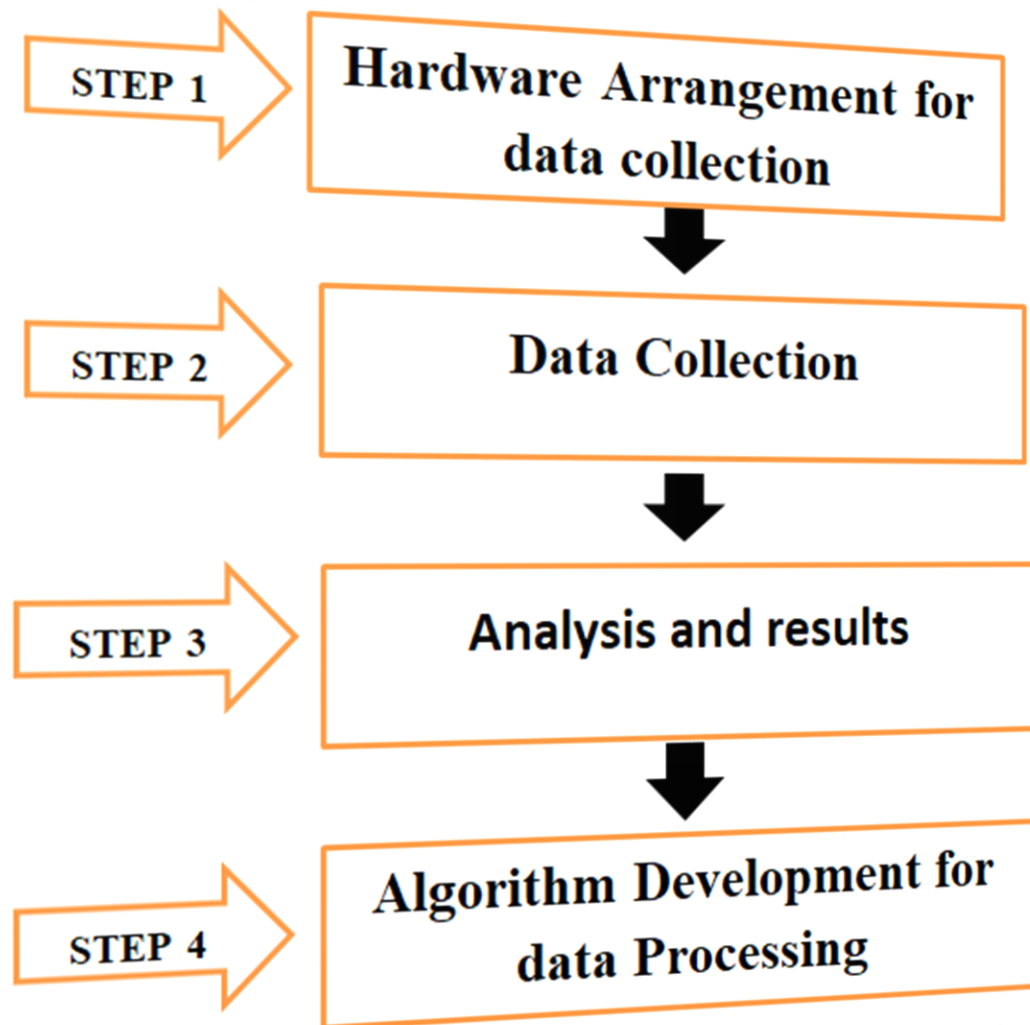


Fig 6.1:Research Work flow Diagram

With the help of my supervisor and a public organization, we performed experiments in the GNSS lab. The first data was taken at an 8-meter distance from the reflecting surface/wall as shown in the diagram and then the next two were for 10 meters. Satellites no 6, 8, 14, 15, 25, 26, 28, 31 were acquired and processed to calculate the delay of the reflected signal.

Please put a diagram of the experimental setup.

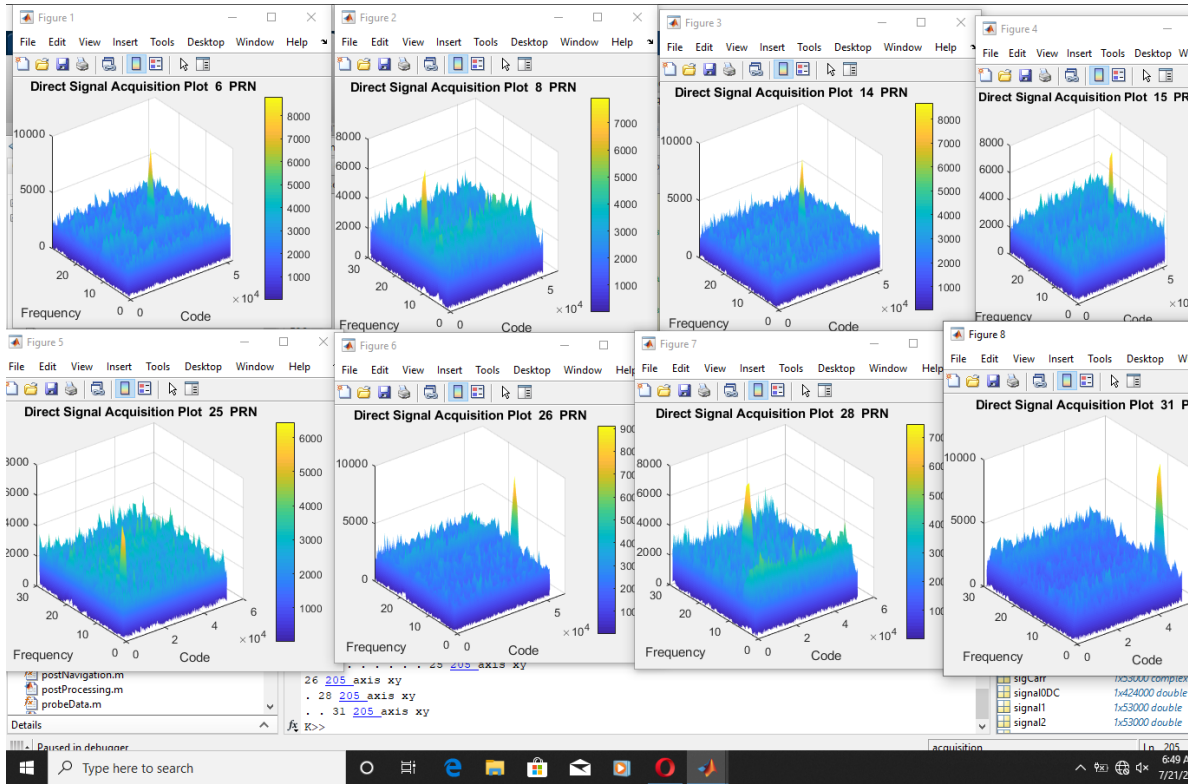


Fig 6.2: Acquired satellites during the experiments

6.1 Signal Processing

The data collected was processed in matlab® using our own developed algorithm for the acquisition and comparison of both direct and reflected signals correlation waveforms. The details of the experimental setup have been shown in the previous chapter. The correlation time can be adjusted from 1ms to 7 ms depending on the strength of the signal during these experiments. The best results were obtained from satellites number 13, 14, 15, and 31.

6.2 Experiment 1

Distance = 8m (from the reflecting surface/wall) and different correlation times (1 ms to 8ms), one chip/code delay is about 5.7 m as explained in section 5.5 (chip resolution)

Distance covered by Reflected signal = 16 (signal round trip)

Time(ms)	1	2	3	4	5	6	7	8
Satellite no								
6	0	0	0	0	1	0	-1	0
8	0	0	0	0	1	0	0	0
13	1	1	1	1	1	1	0	0
14	0	0	0	1	1	1	1	0
15	1	1	1	1	1	1	1	1

26	0	-1	0	0	0	-1	0	0
28	-1	0	-1	-1	-1	-1	-1	-1
31	2	2	2	2	2	1	1	1

Table 6.1 Chip/code delay observed for satellites at different correlation times for 8m distance

6.2.1 Satellite No 31

This satellite showed the best results in terms of chip difference between direct and reflected signals.

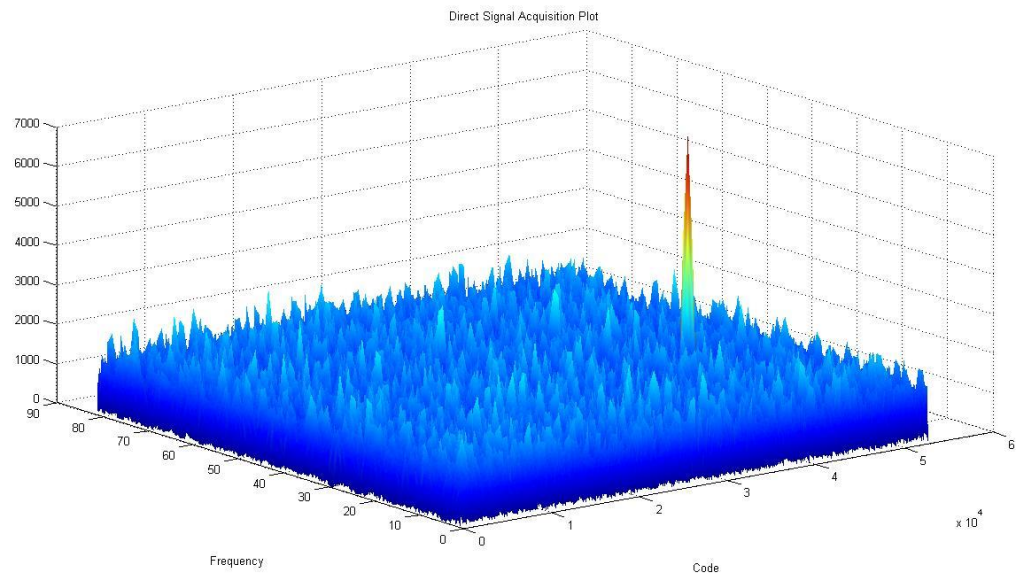


Fig 6.3: Direct Signal Acquisition Plot for Data 1 SV 31

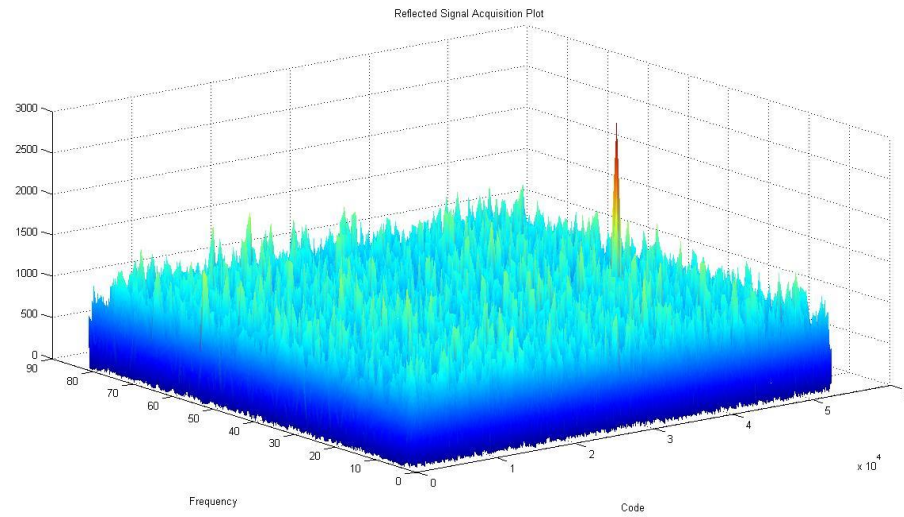


Fig 6.4: Reflected Signal Acquisition Plot Data 1 SV 31

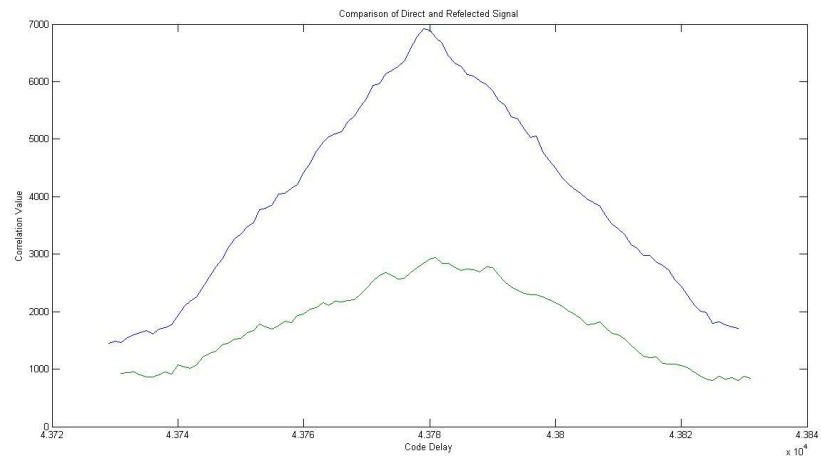


Fig 6.5: Comparison of Direct and Reflected signals Data 1 SV 31

Discussions: As evident from the above diagram the reflected signal is weaker than the direct signal, which is expected and also confirmed that it is indeed the reflected signal. As the bit resolution is 5.7 m so the calculated distance is 11.4.

$16 - 11.4 = 3.6\text{m}$, thus the error lies within Bit resolution

```

psnr_r =
    22.7031

c_freq =
    14625000

ans =
    43829

ans =
    43831

ans =
     2

ans =
     2

fx >>

```

Fig 6.6 Matlab results in terms of a number of chips difference between direct and reflected signals

6.3 Experiment 2

Distance =8m (from the reflecting surface/wall) and different correlation times (1 ms to 8ms), one chip/code delay is about 5.7 m as explained in section 5.5 (chip resolution)

Distance covered by Reflected signal =16 (signal round trip)

Time(ms)	1	2	3	4	5	6	7	8
Satellite Vehicle no								
6	-3	0	-1	-1	-1	0	0	0
8	0	1	1	1	-1	-1	-1	-1
13	0	0	0	0	0	2	2	2
14	0	3	0	0	-1	0	0	0
15	309	1	2	0	2	1	2	2
26	0	-1	0	0	0	0	0	-1
28	-1	0	0	0	0	0	0	0
31	0	0	1	0	0	0	0	0

Table 6.2 Chip delay observed for satellites at different correlation times for 8m distance

6.3.1 Satellite No 13

The above two satellites showed the best results in terms of chip difference between direct and reflected signals.

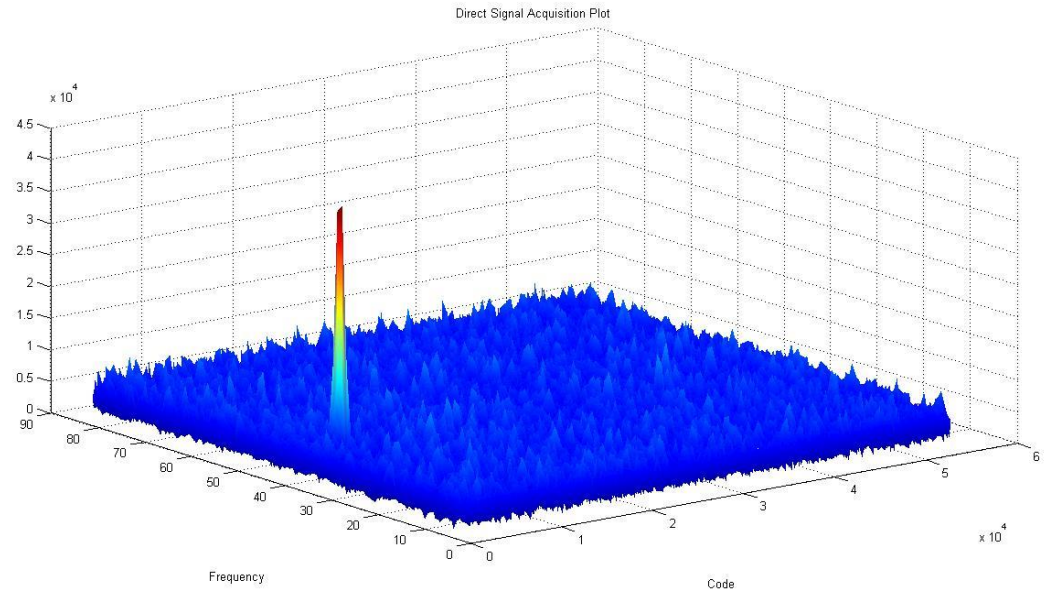


Fig 6.7: Direct Signal Acquisition plot for experiment no 2 SV 13

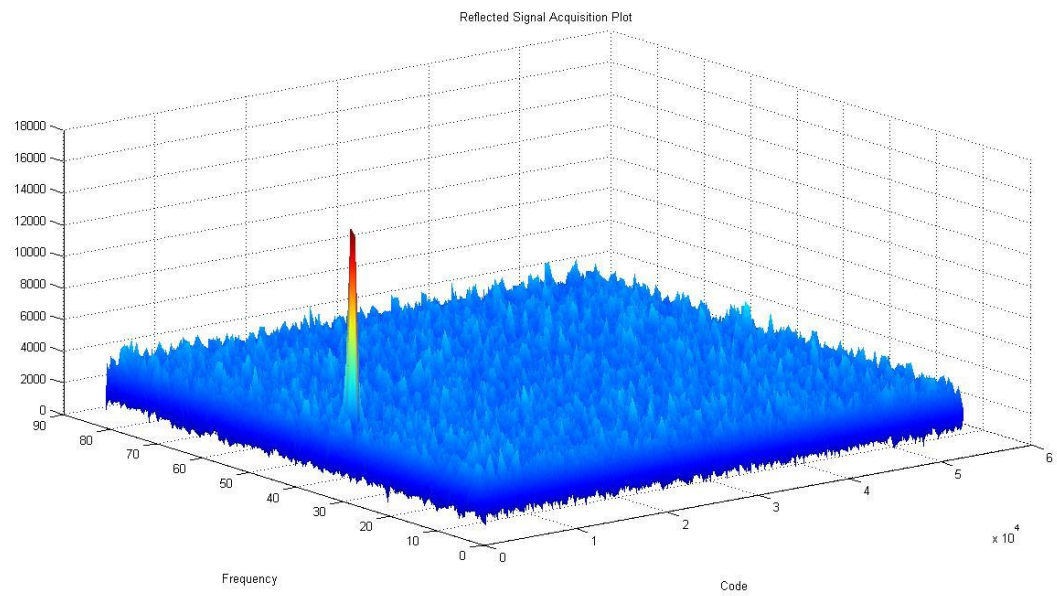


Fig 6.8: Reflected Signal Acquisition plot for experiment no 2 SV 13

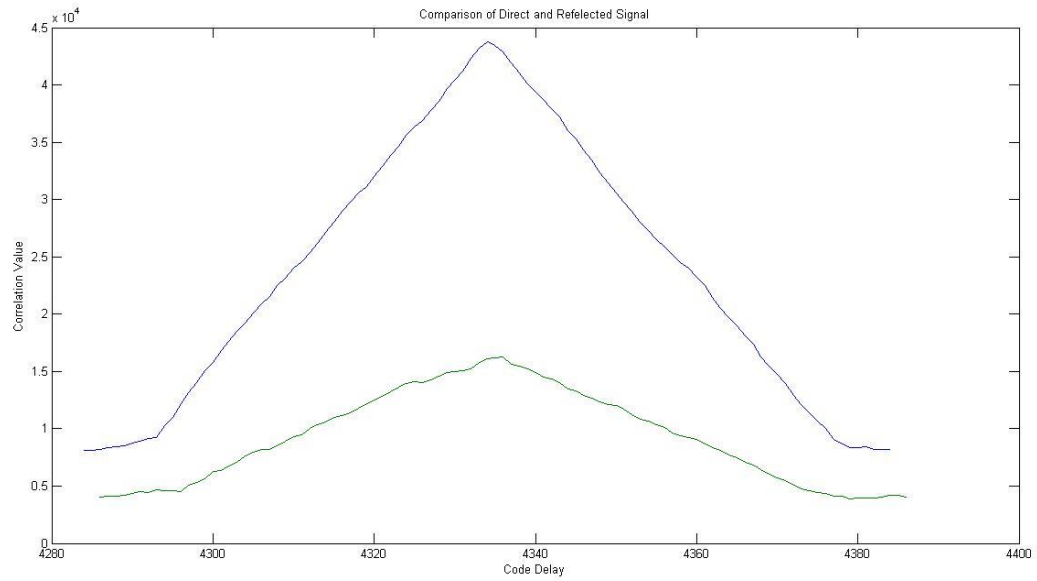


Fig 6.9: Comparison of Direct and Reflected Signal for experiment no 2 SV 13

Discussions: As evident from the above diagram the reflected signal has low power than the direct signal, which is expected and also confirmed that it is indeed the reflected signal. As the bit difference resolution is 5.7 m so the calculated distance is 11.4.

$16 - 11.4 = 3.6\text{m}$, thus the error lies within Bit resolution

6.3.2 Satellite No 15:

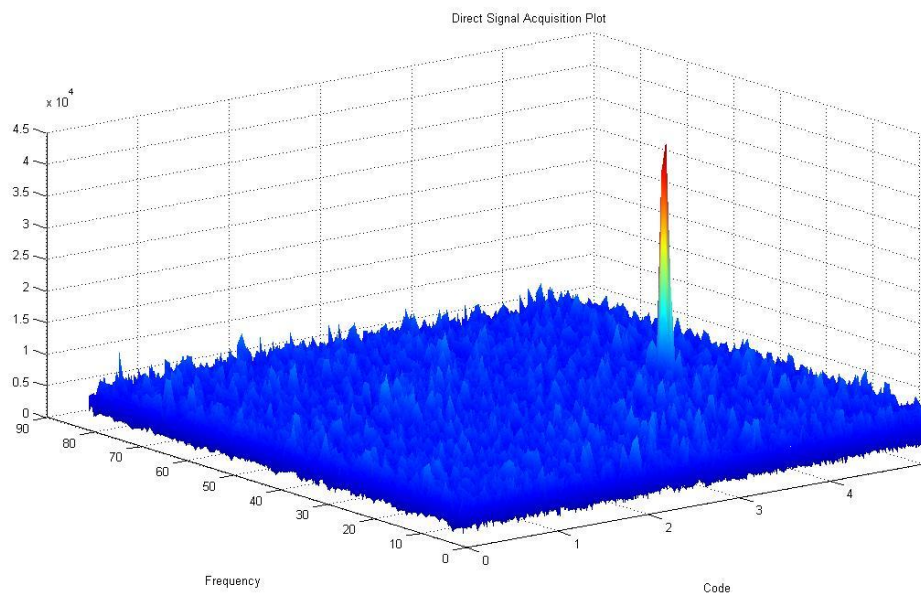


Fig 6.10: Direct Signal Acquisition plot for experiment no 2 SV 15

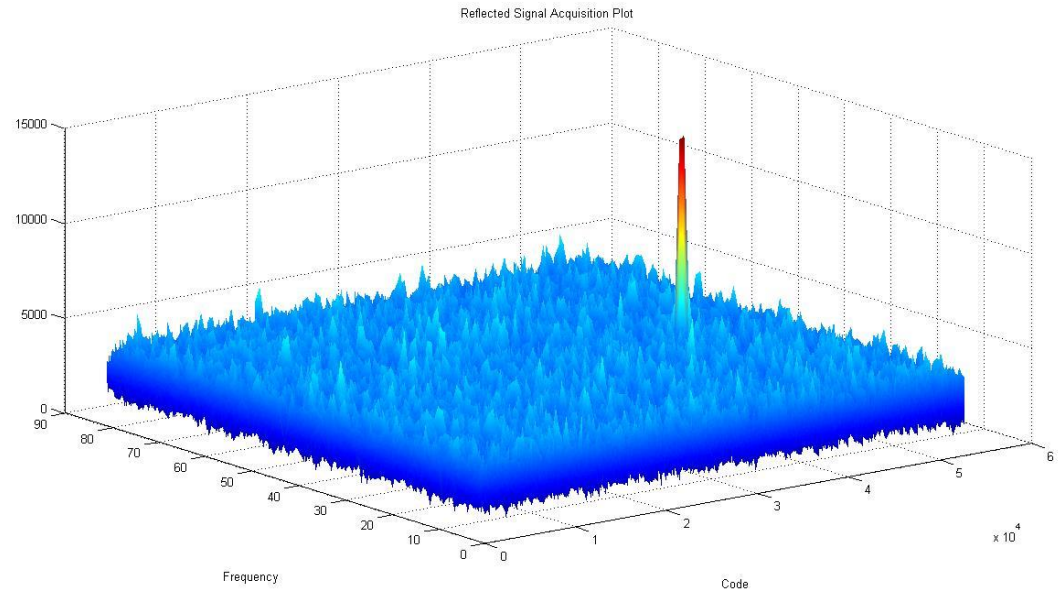


Fig 6.11: Reflected Signal Acquisition plot for experiment no 2 SV 15

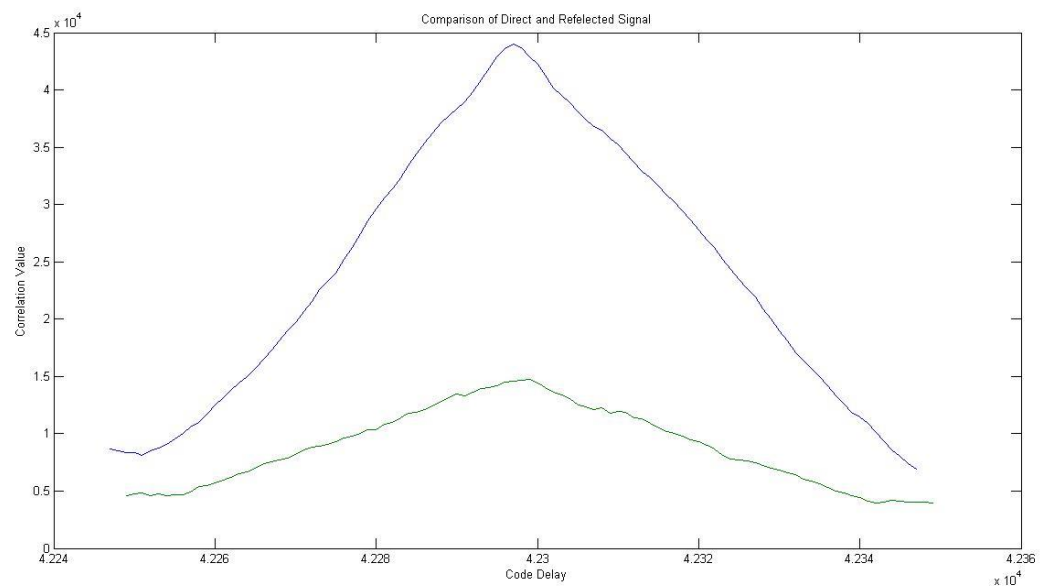


Fig 6.12: Comparison of Direct and reflected signal for experiment no 2 SV o 15

Discussions. As evident from the above diagram the reflected signal has low power than the direct signal, which is expected and also confirmed that it is indeed the reflected signal. As the bit resolution is 5.7 m so the calculated distance is 11.4.

16-11.4=3.6m, thus the error lies within Bit resolution

6.4 Experiment 3

Distance =10m (from the reflecting surface/wall) and different correlation times (1 ms to 8ms), one chip/code delay is about 5.7 m as explained in section 5.5 (chip resolution)

Distance covered by Reflected signal =20 (signal round trip)

Time(ms)	1	2	3	4	5	6	7	8
SV no								
6	3	1	1	1	1	1	1	1
8	1	2	1	1	1	1	1	1
13	1	0	1	0	0	1	1	1
14	1	2	2	1	1	1	1	1
15	4	1	0	2	1	2	2	2
26	0	0	1	0	0	1	1	1
28	2	1	1	1	1	1	1	1
31	1	1	1	1	1	1	1	1

Table 4.3 Chip delay observed for satellites at different correlation times for 10m distance

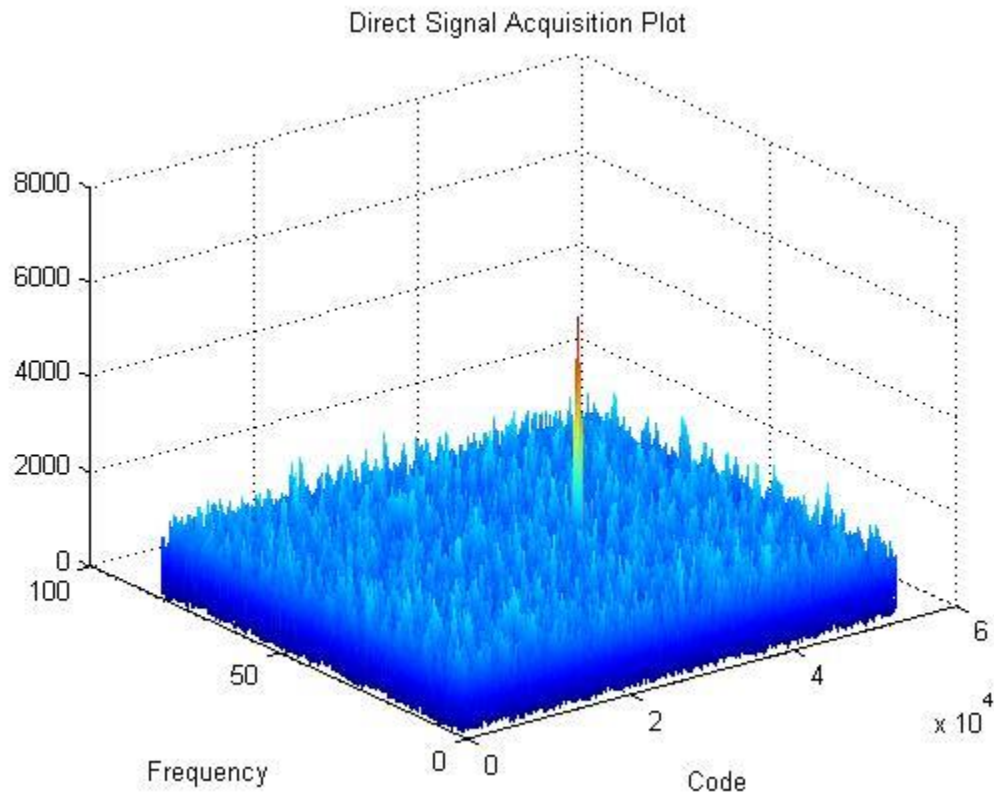


Fig 6.13: Direct signal Acquisition plot for experiment no 3 SV 15

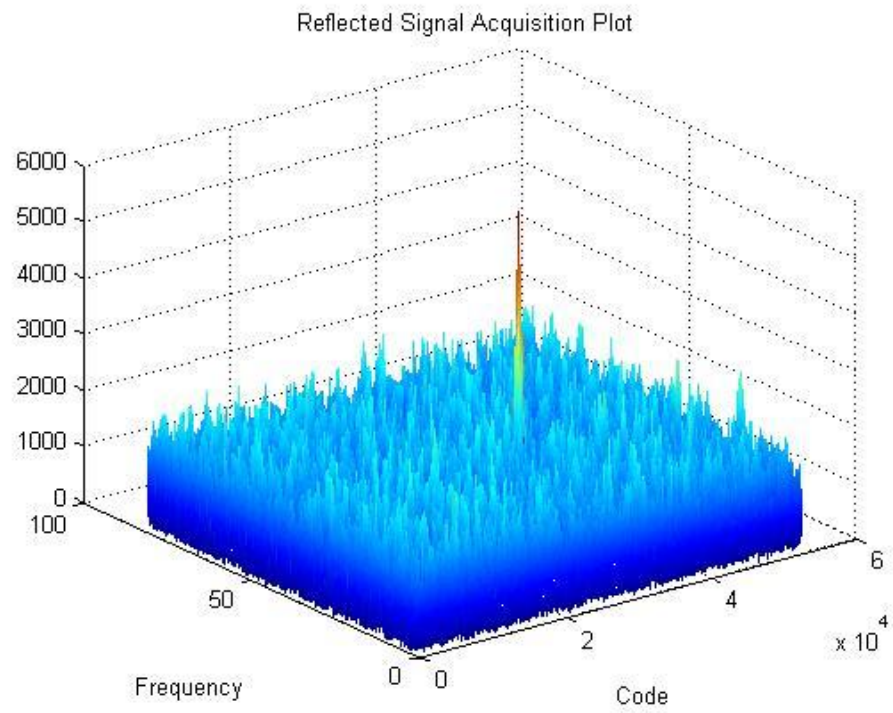


Fig 6.14: Reflected Signal Acquisition plot for experiment no 3 SV no 15

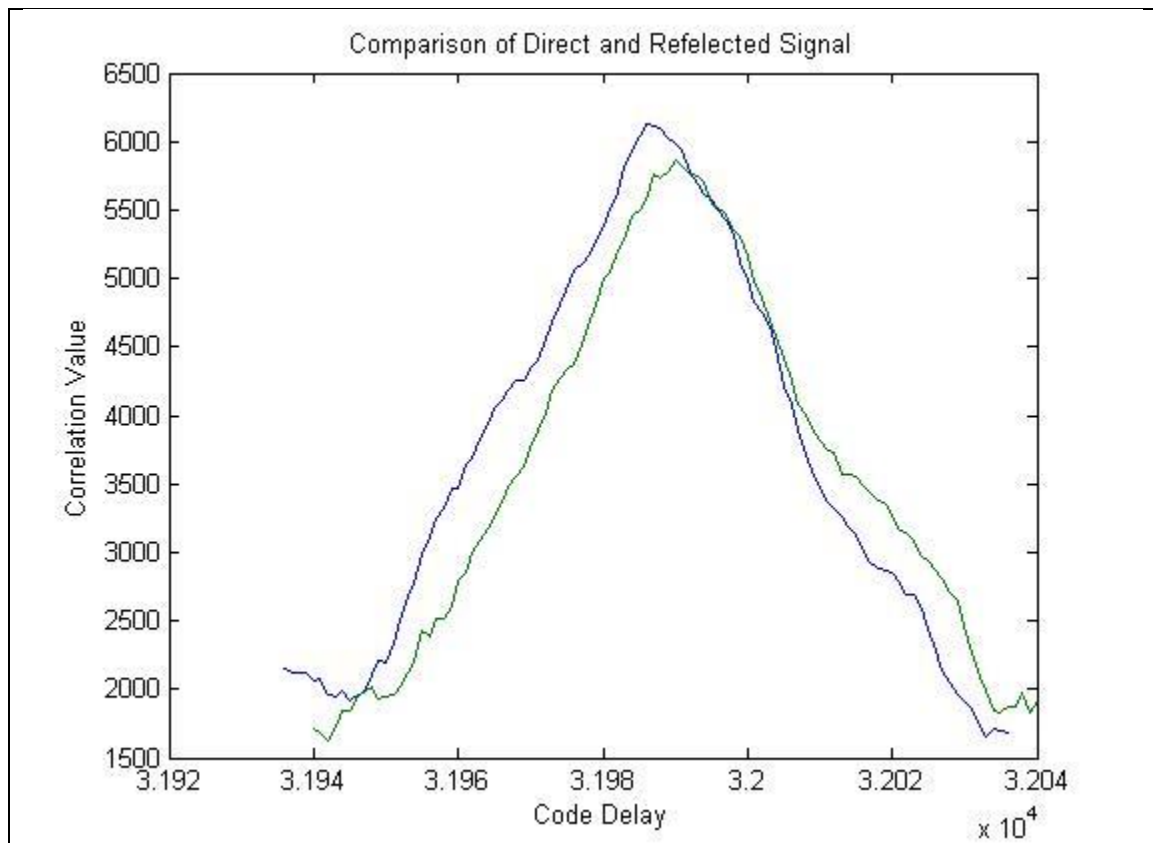


Fig 6.15: Comparison of Direct and reflected signal for experiment no 3 SV 15

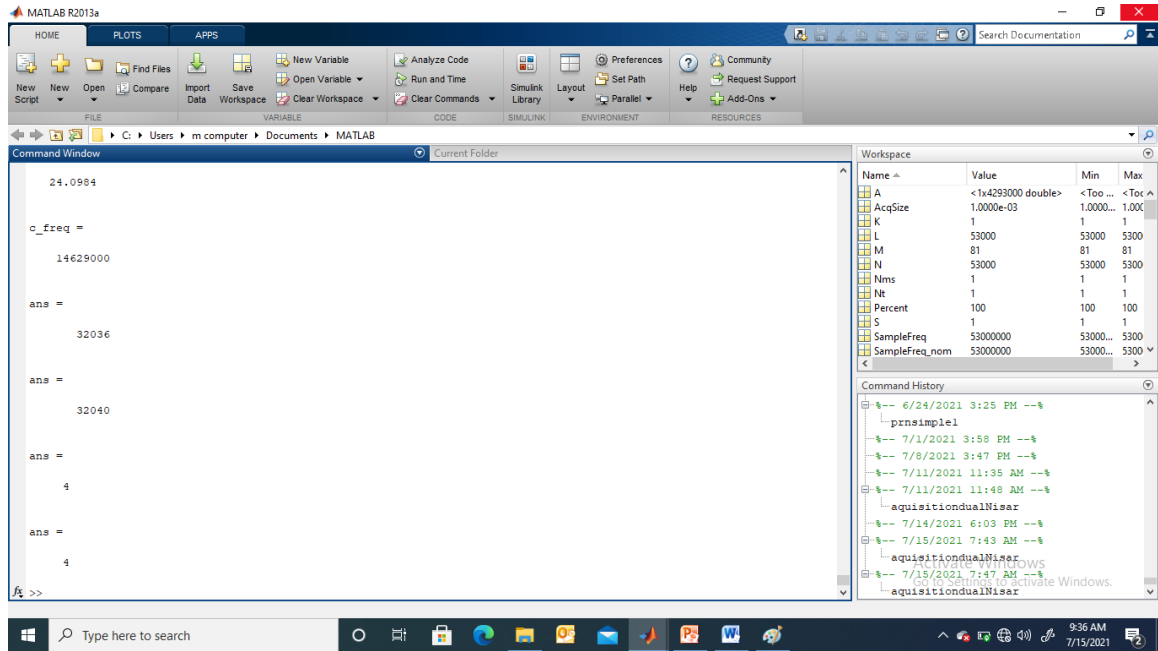


Fig 6.16 Chip difference between Direct and Reflected Signal experiment no 3 SV 15

The actual distance is 20 m and 11.4 is the calculated distance.

Satellite No 6

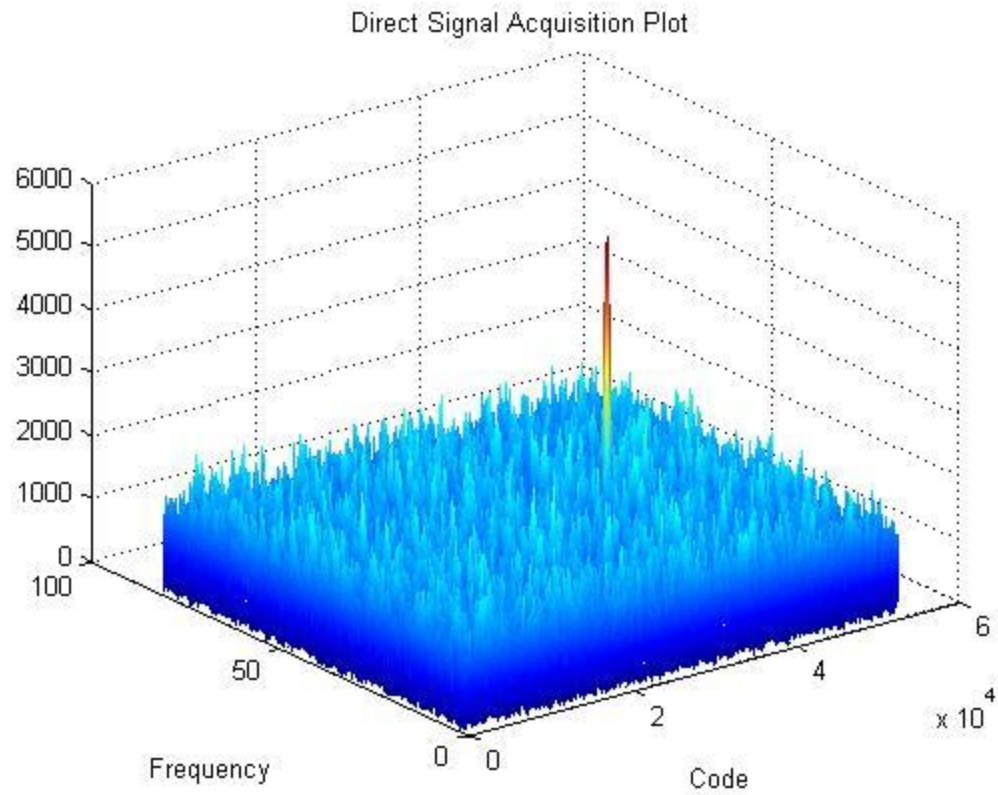


Fig 6.18: Reflected Signal Acquisition plot for experiment no 3 SV 6

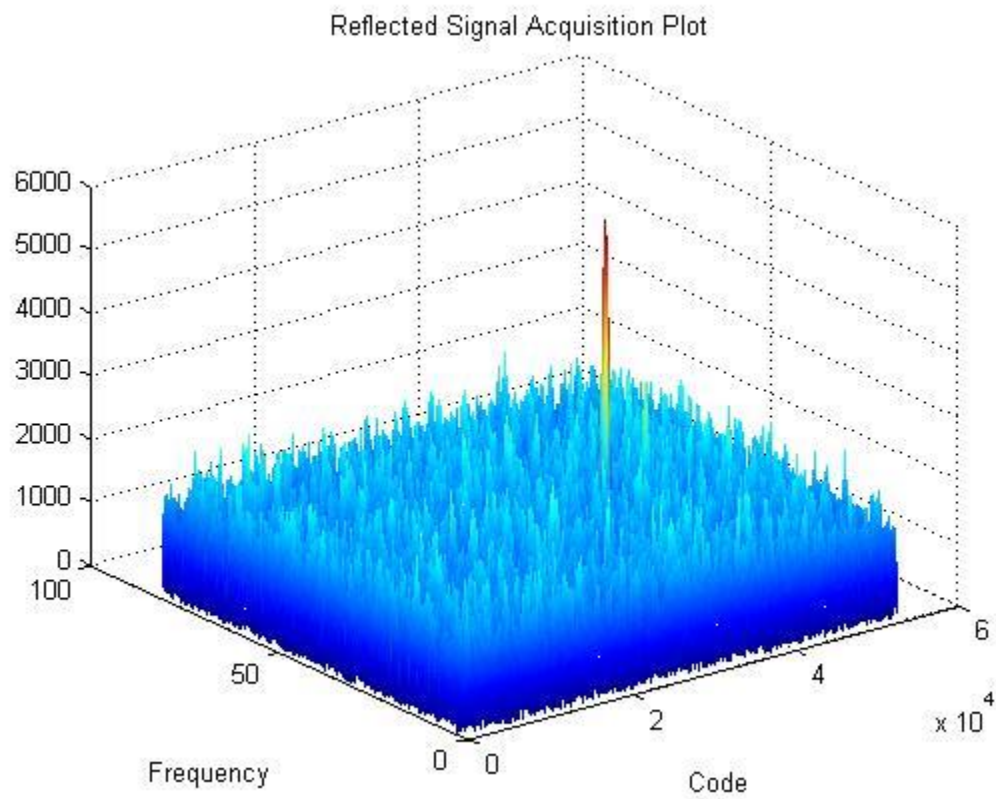


Fig 6.18: Reflected Signal Acquisition plot for experiment no 3 SV 6

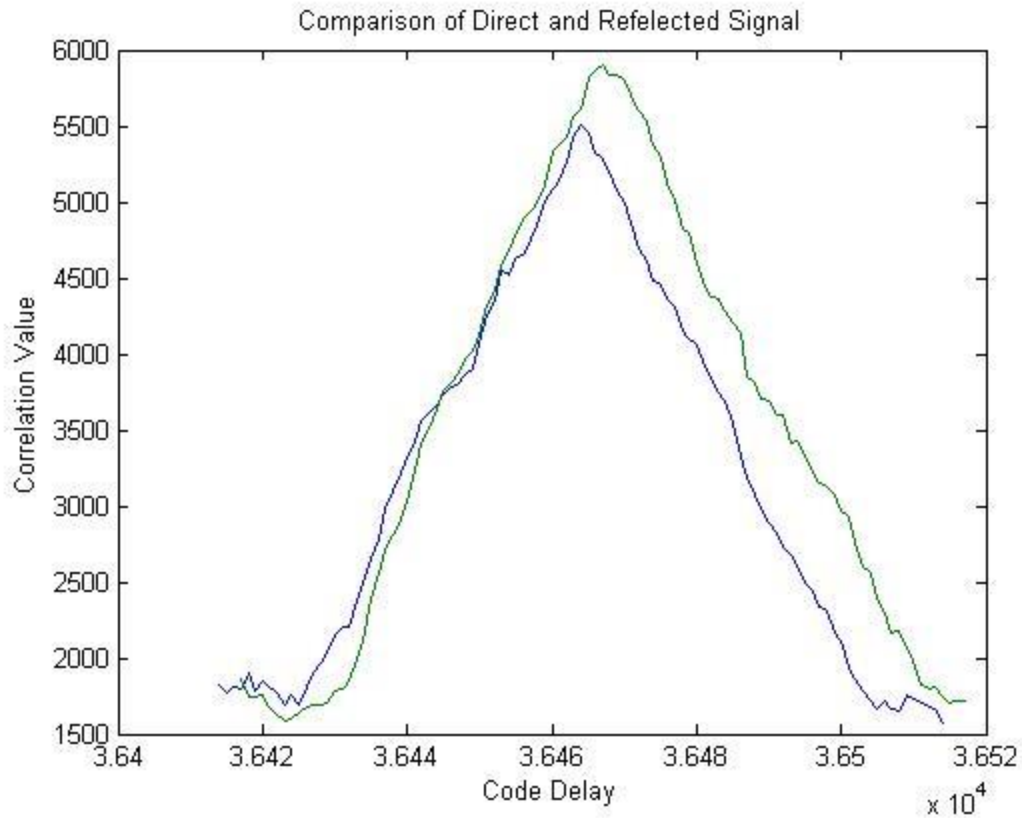


Fig 4.14: Comparison of Direct and reflected signal SV 6

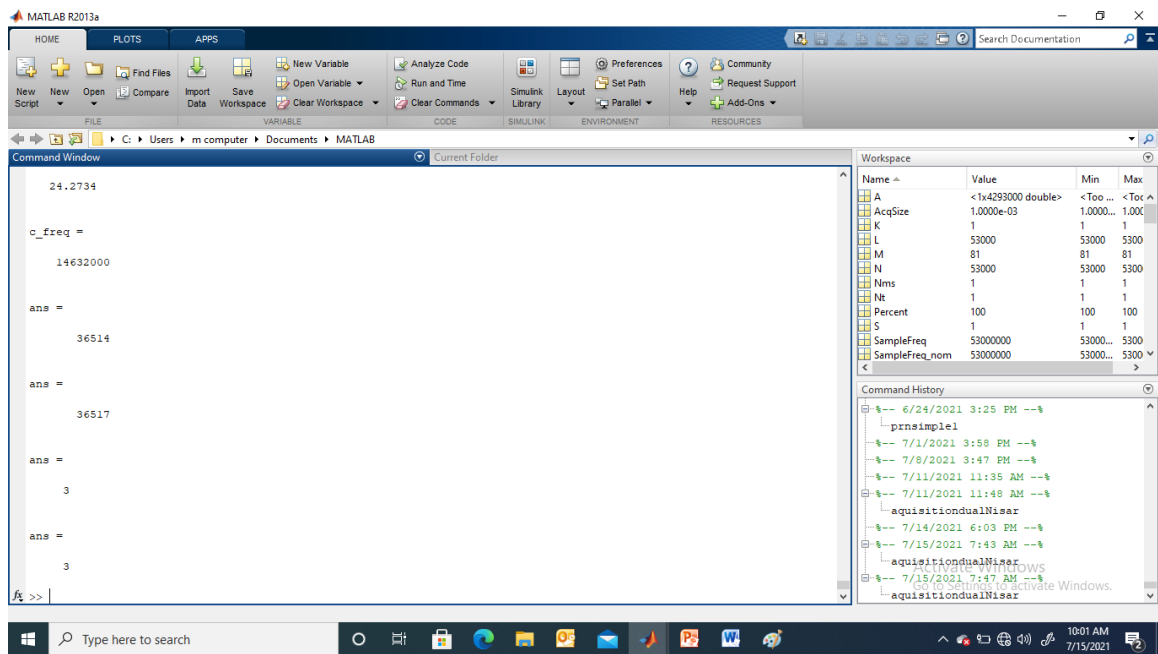


Fig 4.15 Chip difference between Direct and Reflected Signal (3 chip)

Satellite No 26

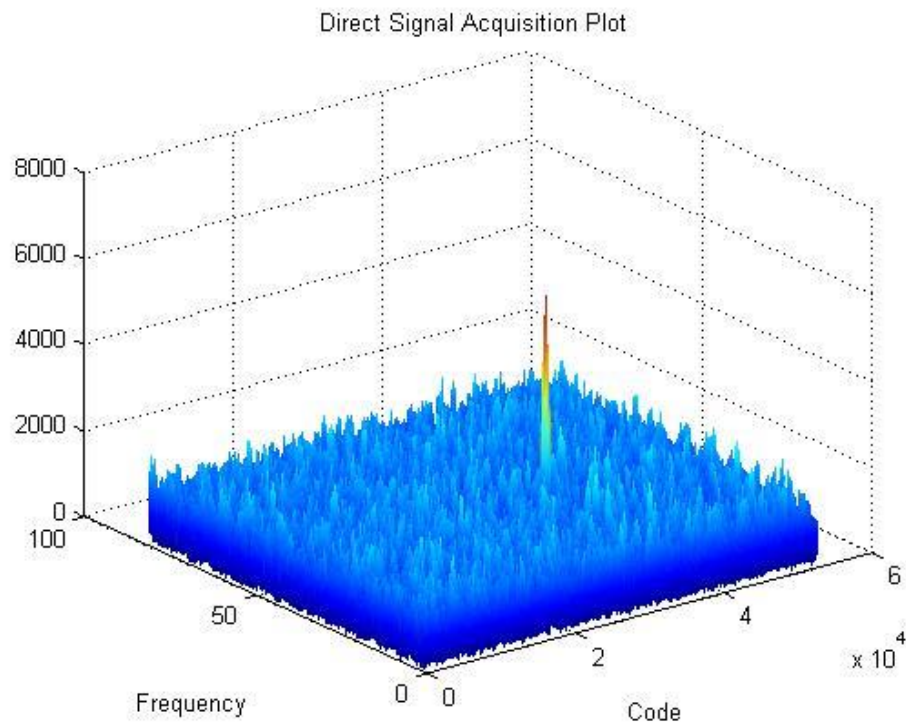


Fig: **Direct signal Acquisition plot for SV no 26**

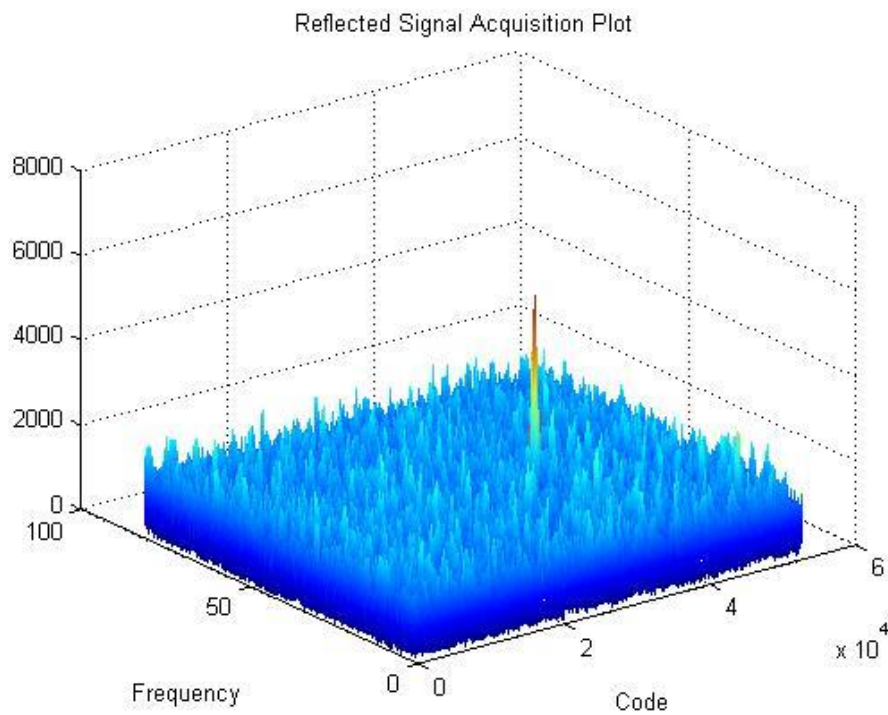


Fig: **Reflected signal Acquisition plot for SV no 26**

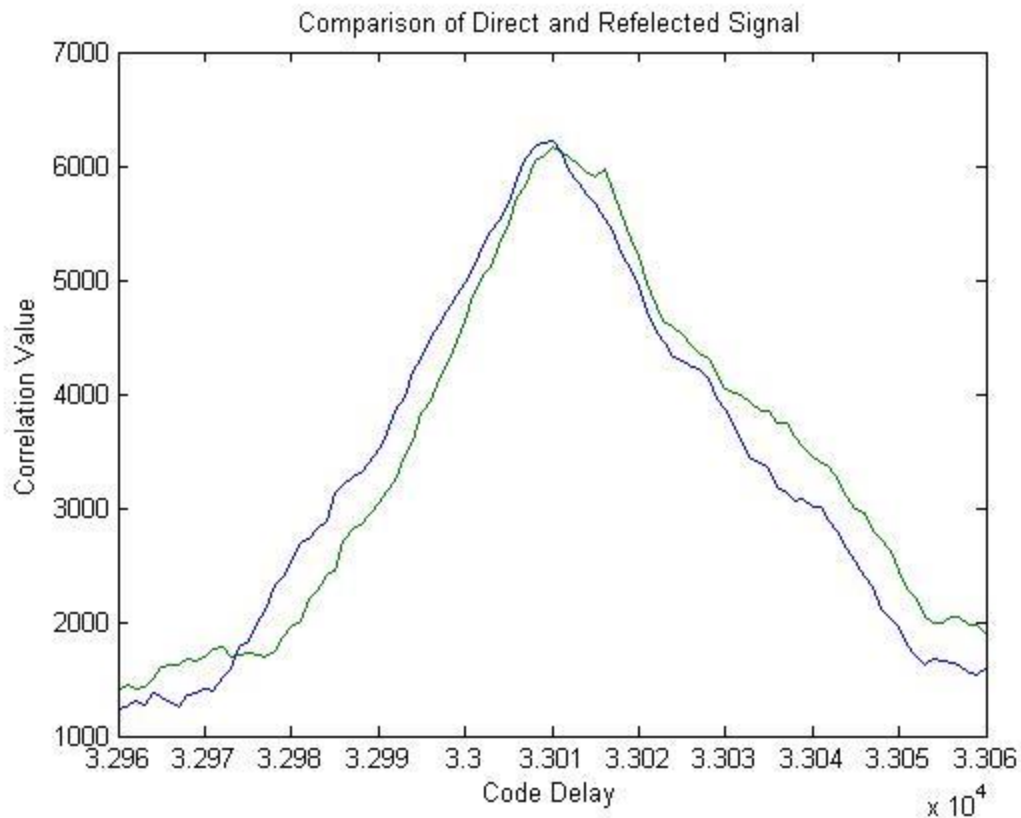


Fig 4.14: Comparison of Direct and reflected signal SV 6

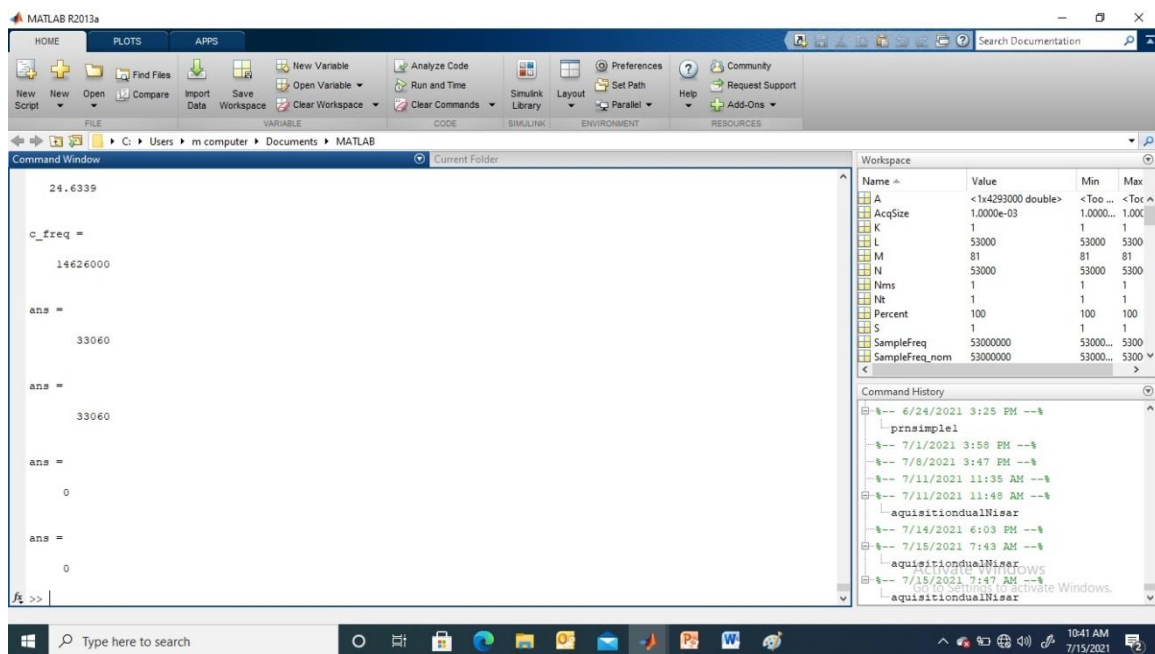


Fig 6.15 Chip difference between Direct and Reflected Signal (0 chip)

7 CONCLUSIONS AND FUTURE WORK

The research work demonstrated an interesting secondary application of GNSS signals in which successful acquisition of reflected GPS signals was carried out for further signal processing. In my research work, I demonstrated an experimental passive GNSS (GPS L1 signal) Radar Altimeter. The experiments have concluded that it is possible to measure distance with the help of reflected GNSS signals.

The experiment was carried out in a Lab, but the GNSS front end can also be installed on a UAV or an aerial platform. The reflected signals can be used to perform height measurements and act as a passive Microwave based Radar Altimeter (RA).

Future work may be included the installation of hardware on the Aerial platform and carry out experiments in the field environment. To improve the resolution the RF data acquisition unit can be designed and prepared with having a higher sampling frequency. Other GNSS signals e.g. BeiDou and Galileo can be utilized for this purpose.

There is ample room for improvement both on the hardware and software side and if more efforts are put in, a successful practical system can be designed and prepared that can be installed on an aerial platform and measure height. A static system can be installed on a bridge or near a dam to continuously measure water levels.

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